

GINAKIT 2

|         |                  |                  |                  |                  |                  |
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GD2 SPECIFIC CHIMERIC ANTIGEN RECEPTOR AND INTERLEUKIN-15 EXPRESSING AUTOLOGOUS NATURAL KILLER T-CELLS TO TREAT CHILDREN WITH NEUROBLASTOMA (GINAKIT2)

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## 1. Patient Eligibility Checklists

### Procurement Checklist

PATIENT ID (MR#): \_\_\_\_\_

PATIENT NAME: \_\_\_\_\_

| The following must be answered "YES" for a patient to be eligible to participate in this study |    |                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| YES                                                                                            | NO | <i>Eligibility criteria for Recruitment</i>                                                                                                                                                                                                 |
|                                                                                                |    | Relapsed or refractory high-risk neuroblastoma                                                                                                                                                                                              |
|                                                                                                |    | Age greater than 1 year and less than 21 years old                                                                                                                                                                                          |
|                                                                                                |    | Life expectancy of at least 12 weeks                                                                                                                                                                                                        |
|                                                                                                |    | Karnofsky/Lansky score 60% or greater ( <b>Appendix A</b> )                                                                                                                                                                                 |
|                                                                                                |    | Absence of HAMA prior to enrollment (only patients who have received prior therapy with murine antibodies)                                                                                                                                  |
|                                                                                                |    | Ability to tolerate leukocyte apheresis                                                                                                                                                                                                     |
|                                                                                                |    | Informed consent and assent (as applicable) obtained from parent/guardian and child                                                                                                                                                         |
|                                                                                                |    | ANC $\geq 500/\mu\text{l}$ #                                                                                                                                                                                                                |
|                                                                                                |    | Platelet count $\geq 20,000/\mu\text{l}$ (may be transfused)                                                                                                                                                                                |
|                                                                                                |    | Pulse Ox $\geq 90\%$ on room air                                                                                                                                                                                                            |
|                                                                                                |    | Serum AST less than 3 times upper limit of normal                                                                                                                                                                                           |
|                                                                                                |    | Total Bilirubin less than 1.5 times upper limit of normal                                                                                                                                                                                   |
|                                                                                                |    | Creatinine $<1.5$ times the upper limit of normal                                                                                                                                                                                           |
|                                                                                                |    | Recovered from the acute toxic effects of all prior chemotherapy based on the enrolling physician's assessment (if some effects of chemotherapy are expected to last long term, patient is eligible if meeting other eligibility criteria). |
|                                                                                                |    | Weight greater than 12 kg                                                                                                                                                                                                                   |
| The following must be answered "NO" for a patient to be eligible to participate in this study  |    |                                                                                                                                                                                                                                             |
|                                                                                                |    | Rapidly progressive disease                                                                                                                                                                                                                 |
|                                                                                                |    | History of hypersensitivity to murine protein-containing products                                                                                                                                                                           |
|                                                                                                |    | Tumor causing airway obstruction                                                                                                                                                                                                            |
|                                                                                                |    | Currently receiving immunosuppressive drugs such as corticosteroids\$, tacrolimus or cyclosporine                                                                                                                                           |
|                                                                                                |    | Severe previous toxicity from cyclophosphamide or fludarabine based on the enrolling physician's assessment                                                                                                                                 |
|                                                                                                |    | HIV infection                                                                                                                                                                                                                               |

#. ANC  $\geq 500$  without the use G-CSF or GM-CSF for at least 48 hrs.

\$. Patients may receive treatment if treated with corticosteroids with dose of less than 0.5 mg/kg/day of prednisone equivalent.

To register patients, contact the CAGT Research Coordinator. To check eligibility, call Dr. Andras Heczey at 832-824-4233.

\_\_\_\_\_  
Signature of Enrolling Physician

\_\_\_\_\_  
Date

## TREATMENT CHECKLIST

PATIENT ID (MR#): \_\_\_\_\_

PATIENT NAME: \_\_\_\_\_

DOSE LEVEL: \_\_\_\_\_

| The following must be answered "YES" for a patient to be eligible to participate in this study |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------|----|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| YES                                                                                            | NO | Value/Date | Eligibility criteria for Treatment*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                |    |            | Relapsed or refractory high-risk neuroblastoma                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                                |    |            | Life expectancy of at least 12 weeks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                |    |            | Age greater than 1 year and less than 21 years old                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                |    |            | Karnofsky/Lansky score 60% or greater ( <b>Appendix A</b> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                |    |            | ANC $\geq 500/\mu\text{l}^\#$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                |    |            | Platelet count $\geq 20,000/\mu\text{l}$ (may be transfused)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                |    |            | Pulse Ox $\geq 90\%$ on room air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                |    |            | Serum AST less than 3 times upper limit of normal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                |    |            | Total Bilirubin less than 1.5 times upper limit of normal                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                |    |            | Creatinine $<1.5$ times the upper limit of normal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                |    |            | Weight greater than 12 kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                |    |            | Recovered from acute toxic effects of all prior chemotherapy based on the enrolling physician's assessment (if some effects of chemotherapy are expected to last long term, patient is eligible if meeting other eligibility criteria and expected to tolerate lymphodepletion).                                                                                                                                                                                                                                                 |
|                                                                                                |    |            | Absence of HAMA prior to enrollment (only patients who have received prior therapy with murine antibodies)                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                |    |            | Autologous, transduced peripheral blood NKT-cells with $\geq 20\%$ expression of GD2-CAR                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                                |    |            | Informed consent and assent (as applicable) obtained from parent/guardian and child.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| The following must be answered "NO" for a patient to be eligible to participate in this study  |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                |    |            | Rapidly progressive disease                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                |    |            | Currently receiving any investigational drugs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                |    |            | History of hypersensitivity to murine protein-containing products                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                |    |            | Cardiomegaly or bilateral pulmonary infiltrates on chest radiograph or CT. However, patients with cardiomegaly on imaging may be enrolled if they have an assessment of cardiac function (i.e., ECHO or MUGA) within 3 weeks of starting protocol therapy that is within normal limits. Additionally, patients with bilateral pulmonary infiltrates on imaging may be enrolled if the lesions are not consistent with active neuroblastoma (i.e., negative on functional imaging with PET or MIBG, or by pathologic assessment). |
|                                                                                                |    |            | Tumor potentially causing airway obstruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                |    |            | Pregnancy or lactation / will not use birth control methods                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                |    |            | Currently receiving immunosuppressive drugs such as corticosteroids <sup>§</sup> , tacrolimus or cyclosporine                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                |    |            | Severe previous toxicity from cyclophosphamide or fludarabine based on the enrolling physician's assessment                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                |    |            | HIV infection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

\*: All labs must be collected within 10 days prior to initiation of study-related treatment (except for verification of GD2 transduction)

<sup>#</sup>: ANC  $\geq 500/\mu\text{l}$  without the use G-CSF or GM-CSF for at least 48 hrs.

<sup>§</sup>: Patients may receive treatment if treated with corticosteroids with dose of less than 0.5 mg/kg/day of prednisone equivalent.

To register patients for NKT infusion, contact the CAGT Research Coordinator. To check treatment eligibility, call Dr. Andras Heczey at 832-824-4233.

\_\_\_\_\_  
Signature of Enrolling Physician

\_\_\_\_\_  
Date

## 2. Objectives

- 2.1. **Primary objective:** To determine the maximum tolerated dose (MTD) and safety profile of autologous NKTs expressing a 2<sup>nd</sup>-generation GD2-specific chimeric antigen receptor that incorporates the co-stimulatory endodomain of CD28 in addition to the CD3 zeta chain as well as Interleukin-15 (hereafter referred to as G28z.15 NKTs) administered to patients with relapsed or refractory neuroblastoma. We will assess:
  - 2.1.1. The MTD of G28z.15 NKTs adoptively transferred to patients with relapsed or refractory neuroblastoma,
  - 2.1.2. The safety profile of G28z.15 NKTs adoptively transferred to patients with relapsed/refractory neuroblastoma.
- 2.2. **Secondary objective:** To evaluate the anti-tumor response of autologous G28z.15 NKT cells in patients with relapsed/refractory neuroblastoma. We will assess:
  - 2.2.1. Evaluate residual disease sites for change of dimensions of tumor mass by 3D imaging (MRI or CT) after the administration of autologous G28z.15 NKT cells.
  - 2.2.2. Determine the number of detectable metaiodobenzylguanidine (MIBG) avid disease or bone scan positive sites in response to the administration of autologous G28z.15 NKT cells.
  - 2.2.3. Assess bone marrow clearance among children with bone marrow involvement at the time of enrollment after the administration of autologous G28z.15 NKT cells.
- 2.3. **Additional Objective:** To evaluate the immunologic response of autologous G28z.15 NKT cells in patients with relapsed/refractory neuroblastoma. We will assess:
  - 2.3.1. The expansion and functional persistence of G28z.15 NKTs in the peripheral blood of patients using transgene detection by quantitative real-time PCR and flow cytometry.
  - 2.3.2. The frequency of NKT target cells: CD1d+CD33+CD14+CD163+/-myelomonocytic cells in blood and in tumor biopsies and BM aspirates (when available).
  - 2.3.3. The frequency and activation status (CD69) of NK cells (CD3-CD56+),
  - 2.3.4. The sequential changes in patients' serum cytokine and chemokine levels after G28z.15 NKT infusion.
  - 2.3.5. The changes in CAR NKTs and cancer cells post-infusion in tumor biopsies.

## 3. Background

- 3.1. **Neuroblastoma**, the most common extracranial solid tumor of childhood, affects approximately 600 children a year in the United States. When the tumor occurs in infants (children <1 year of age), it is frequently presents at lower stage and responds

well to therapy (1). Even disseminated disease can be eradicated in about 75% of infants and, indeed, may undergo spontaneous remission (1, 2). In older children (>1 year of age), the prognosis is far worse. Although patients with localized disease may still be cured by conventional therapy, a significant number of patients with disseminated tumor can be expected to relapse within 3 years, and virtually none of this subgroup will become long-term survivors (3). Over the past decades, attempts to improve the outcome of advanced neuroblastoma have focused on greater intensification of the induction and consolidation phases of chemo-radiotherapy, with or without stem cell rescue (4-7). Despite improvements in remission rates, long-term survival remains poor and those who survive may have severe long-term effects from the intensified treatment regimens (8). The failure to significantly improve outcomes has led to a resurgence of interest in alternative methods of disease eradication of which immune modulation is one particular option (9, 10).

3.2. **Immunotherapy of Neuroblastoma:** While our current study is the first to propose the use of NKT cells for immunotherapy of neuroblastoma, the concept of exploiting the immune system to eradicate this disease has had a long history (2). Because neuroblastoma is derived from embryonic neuroectoderm, it expresses antigens not widely detected in non-embryonic tissues (11, 12) and may overexpress other cellular antigens (3). Finally, in animals and in man, neuroblastoma cells are susceptible to cytotoxic immune-effector mechanisms both in vitro and in vivo (13, 14). In the past, clinical immunotherapy for neuroblastoma has taken many forms, most of which may be loosely classified as vaccine/ cytokines or the use of therapeutic monoclonal antibodies.

3.2.1. One of the earliest attempts to use immunotherapy was to immunize patients with disrupted tumor mixed with bacillus Calmette-Guérin (BCG). These crude mixtures produced severe local reactions, but did lead to detectable, albeit transient, tumor responses (1982 protocol developed at St. Jude Children's Research Hospital, unpublished results). Recombinant cytokines such as IL-2, which stimulate the immune system,(15) or agents such as IFN- $\gamma$ , which may render neuroblastoma cells more immunogenic to cytotoxic T cells by increasing their expression of class I MHC molecules have also been tried (16). More recently, gene-modified tumor vaccines have been used. This approach is based on murine studies showing that tumor cell lines transfected with cytokine genes show a substantial increase in their immunogenicity (17). These tumor vaccines enhanced the immune response sufficiently to reject the modified tumor cells, as well as non-transduced tumor cells administered concomitantly or shortly thereafter. Our own group has studied the effects of expressing the IL-2 gene alone or in combination with lymphotactin in autologous or allogeneic neuroblasts, which were then injected into children with advanced neuroblastoma (18-20). In patients receiving IL-2 alone, significant tumor responses were seen, including 3 complete remissions among 28 patients treated to date. These patients have also made antibodies that react with autologous and allogeneic neuroblastoma cells in cell surface immunofluorescence studies (18).

3.2.2. The most widely used immunotherapy for neuroblastoma consists of murine or human-murine chimeric monoclonal antibodies directed against GD2, a

disialoganglioside expressed in tumors of neuroectodermal origin. There have been Phase I, II, and III studies using these antibodies, and the reported data from the Children's Oncology Group Phase III study revealed a significant improvement in the 2-year event free and overall survival rate with the addition of GD2 antibody therapy (in combination with IL-2 and GM-CSF) to standard therapy; however, pain, blood pressure abnormalities, fever, rash and neurologic toxicities were major adverse effects associated with this therapy (21). The GD2-specific map is now FDA approved for the upfront therapy of high-risk neuroblastoma for patients in maintenance. Other studies have coupled these antibodies to toxins or to radionuclides with broadly promising results (22-25). CSF and IL-2 associated with additional toxicities including capillary leak syndrome, fever, bronchoconstriction etc. (1-4). More recently, dinutuximab in combination with irinotecan, temozolomide and GM-CSF (DITG cohort) was tested by the COG in children with relapsed or refractory neuroblastoma (26). Among the 17 patients assigned to the DITG cohort, five patients achieved complete response, four had partial response, four had stable disease, three had progressive disease and one patient was not evaluated (26). This study highlights the potential for GD2-targeting for patients with relapsed or refractory neuroblastoma.

- 3.3. **Immunotherapy with chimeric antigen receptors:** Chimeric antigen receptors are generated first by joining the heavy and light chain variable regions of a monoclonal antibody with a linker to form a single-chain Fv (scFv) molecule. This scFv is then attached to the transmembrane and cytoplasmic portion of T Cell Receptors (TCR)  $\zeta$  chain via a flexible hinge region. Engagement of the extracellular scFv of the chimeric receptor results in tyrosine phosphorylation of immune-receptor activation motifs present in the cytoplasmic domain, initiating T cell signaling to the nucleus. Human T lymphocytes genetically engineered to express these recombinant receptor genes have exhibited specific lysis via the perforin/granzyme pathways, as well as cytokine secretion upon exposure to tumor cells expressing the cognate target antigen (27). Engagement of single T cell or Fc receptor chains suffices to induce cellular activation and proliferation (28-31). Chimeric receptor transduced effector cells have numerous advantages over immunotherapies based on monoclonal antibodies or non-CAR expressing lymphocytes alone. Since there is no need to select and expand tumor-associated antigen specific cells from scanty precursors, large populations of antigen-redirected effector lymphocytes can be obtained in a matter of weeks. Moreover, chimeric antigen receptors are MHC-unrestricted, so that neuroblastoma tumor escape, by downregulation of HLA class I molecules or defects in antigen processing are bypassed. The presence of chimeric antigen mediated effector function may be more likely to produce tumor cell lysis than humoral immune responses alone. The perforin/granzyme killing mechanism may be effective against cells that are relatively resistant to antibody and complement, while cytokine secretion upon effector cell activation by tumor antigen recruits additional components of the immune system, amplifying the antitumor immune response. Furthermore, unlike intact antibodies, effector lymphocyte like NKTs can migrate through microvascular walls, extravasate and penetrate the core of solid tumors to exert their cytolytic activity. Finally, a single effector lymphocyte cell can sequentially kill a multiplicity of target cells over a prolonged period of time.

Chimeric antigen receptors (CARs) are fusion proteins consisting of a single chain variable fragment (scFv) derived from a tumor antigen-specific antibody, linked to intracellular signaling domains via spacer and transmembrane regions. 1<sup>st</sup> generation CARs contained only a single intracellular signaling domain most commonly from the CD3 $\zeta$  chain, while 2<sup>nd</sup>- and 3<sup>rd</sup>-generation CARs incorporated signaling endodomains from costimulatory receptors (CD28, 4-1BB, OX40, ICOS etc.) (32).

3.3.1. Experience with T cells expressing a 1<sup>st</sup> generation GD2-specific CAR in patients with relapsed/refractory neuroblastoma: We combined the promising clinical cellular and antibody mediated immunotherapies described above by generating cytotoxic T cells expressing 1<sup>st</sup> generation chimeric single chain antibodies (CARs) directed against GD2. In this study, 1<sup>st</sup> generation, GD2-specific CAR expressing activated T cells (ATCs) and Epstein-Barr virus-specific cytotoxic T cells (EBV-CTLs) were given to 19 patients with a history of

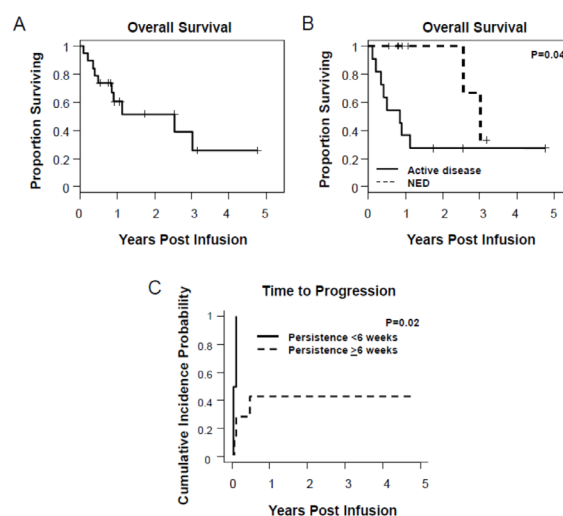
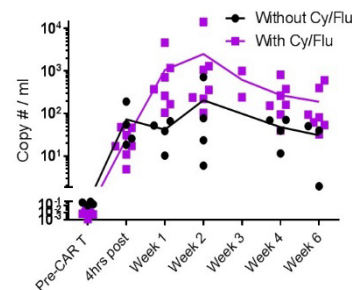


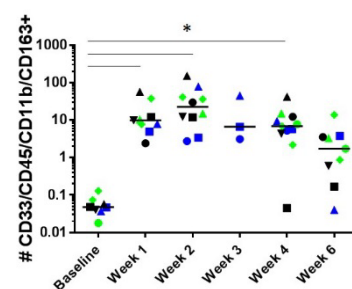
Figure 1

relapsed/refractory neuroblastoma. Five of 11 patients with active disease showed tumor responses or necrosis, and in 3 of these individuals, the responses were complete (33). Of note, progression free survival (**Fig 1 A and B**) was strongly correlated with persistence of GD2-CAR T cells beyond 6 weeks (**Fig 1C**), even though these cells were only present long-term at low levels. EBV-CTLs persisted longer and prolonged GD2-specific CAR expressing T cell persistence within the peripheral blood was associated with the presence of both CD4<sup>+</sup> T cells and central memory T cells within the infused product (33, 34).

3.3.2. Evaluation of T cells expressing a 3<sup>rd</sup>-generation GD2-CAR in patients with relapsed/refractory neuroblastoma: In our previous study, targeting GD2 with a 1<sup>st</sup> generation chimeric antigen receptor expressed in activated T cells or Epstein-Barr-specific T cells found that the approach was safe, and though it could induce antitumor responses, poor expansion and long-term CAR T cell persistence limited efficacy. To overcome this challenge, we developed a 3<sup>rd</sup>-generation GD2-CAR incorporating CD28 and OX40 costimulatory endodomains (GD2-CAR3) that significantly improved persistence and antitumor activity of T cells in preclinical models. Thus, we hypothesized that GD2-CAR3 T cells would be safe and effective in patients with relapsed/refractory NB. This hypothesis was tested in a Phase 1 study at Texas Children's Hospital enrolling relapsed/refractory NB patients with active disease in three cohorts (IND 15609, [GRAIN] NCT01822652). The study was conducted with adaptive trial design where Cohort 1 received GD2-CAR3 T cells alone, cohort 2 received GD2-CAR3 T cells plus conditioning with cyclophosphamide/fludarabine (Cy/flu) and cohort 3 was treated with CAR T cells, Cy/flu and a PD-1 inhibitor. GD2-CAR3 T cell dose escalation ( $10^7/m^2$  to  $1.5 \times 10^8/m^2$ ) was guided by the modified Continual Reassessment Method. We treated 11 relapsed/refractory NB patients with GD2-CAR3 T cells and all patients were included in primary and safety analyses. The infusions were safe and no dose-limiting toxicities occurred. Measurable GD2-CAR3 T cells expansion was seen in cohort 1; importantly, Cy/flu increased circulating IL-15 ( $p=0.003$ ) and improved GD2-CAR3 T cells expansion ( $p=0.03$ ) (**Fig 2**). Our preclinical results showed that NB cells can upregulate PD-L1 and may inhibit GD2-CAR3 T cells through PD-1; however, PD-1 inhibition with pembrolizumab did not further modify expansion or persistence of CAR T cells. Although antitumor responses at 6 weeks were modest, patients in lymphodepletion cohorts 2 and 3 survived longer and 2 subjects (both in cohort 3) had complete responses following salvage therapy. All cohorts showed a striking expansion of CD45/CD33/CD11b/CD163 "inhibitory" M2 macrophage-like myeloid cells (**Fig 3**, change to baseline,  $p=0.0126$ ). Thus, GD2-CAR3 T cells are safe and Cy/flu can further increase their expansion. The unexpected robust expansion of M2-macrophage-like myeloid cells may have contributed to the modest early antitumor responses and we expect that countermeasures to pre-existing and evolving tumor immune evasion will produce both a more immediate tumor response and a superior outcome in future clinical studies.



**Figure 2: Expansion of GD2-CAR3 T cells:** Peripheral blood CAR T cell levels were evaluated by qPCR at indicated time points.



**Figure 3: Expansion of M2-like myeloid cells:** Peripheral blood monocytic cells were evaluated by FACS analysis at indicated time points.

- 3.4. **Role of Natural Killer T-cells in neuroblastoma:** Although the majority of solid tumors, including neuroblastoma, are CD1d-negative, numerous models of cancer have shown the anti-tumor potential of Natural Killer T cells (NKTs) (35). Our group observed that NKT-cell infiltration of primary neuroblastoma is associated with good outcome (36), a finding that has since been extended to other malignancies (37-39). In contrast to many adults with cancer (40, 41), NKTs are numerically and functionally intact in the peripheral blood of children with neuroblastoma, even when they have stage-4 disease (36). We showed that NKTs traffic to NB tissues in response to tumor-derived chemokines CCL2 and CCL20 and actively co-localize with tumor-associated macrophages (TAMs) in primary tumors. Upon recognition of CD1d-bound tumor-derived glycolipids, NKTs can specifically kill TAMs or their monocytic precursors (42). TAMs are phenotypically similar to M2 macrophage-like myeloid cells detected in our previous study of GD2-CAR3 T cells in patients with relapsed / refractory neuroblastoma. Since TAMs provide critical stromal support for tumor cell growth in neuroblastoma and many other types of cancer, NKT cell-mediated killing or inhibition of TAMs explains how NKTs may indirectly affect tumor growth. Without direct attack on neuroblasts, this effect, however, can only transiently slow down tumor progression in a metastatic neuroblastoma model in mice (43). In this study, we will exploit this newly discovered anti-tumor defense mechanism to develop NKT cell-based immunotherapy against neuroblastoma by genetically engineering NKTs to target neuroblasts directly while maintaining their ability to kill TAMs.
- 3.5. **Clinical use of NKTs and their ligands in cancer immunotherapy.** The first NKT ligand discovered was  $\alpha$ -galactosylceramide ( $\alpha$ GalCer, KRN7000, Kirin Brewery Co, Japan), which was derived from the marine sponge *Agelas mauritanus*. Although it demonstrated potent anti-tumor properties in mice (35), administration of KRN7000 to human subjects with cancer showed scant biological activity and failed to expand endogenous NKTs (44). Subsequent clinical trials infused monocyte-derived immature or mature DCs pulsed with KRN7000 (45-47). In the latter study, NKTs were expanded *in vivo*, and increased levels of NKTs in a few patients were detected for up to 6 months after infusion. In another, phase-I/II clinical trial, Motohashi et al adoptively transferred NKTs to patients with advanced and recurrent non-small cell lung cancer (HNSCC)(48). The authors detected an increased number of IFN $\gamma$ -producing NKTs in the peripheral blood (PB) of ~ 60% patients and these patients had double the median survival time of non-responders. However, the causal relationship between NKT-cell biological response and the clinical outcome has not been established. In another clinical trial, NKTs were administered by intra-arterial infusion into tumor-feeding arteries in eight patients with locally recurrent and refractory HNSCC. This produced rapid tumor reduction (partial response) in one patient and stable disease in four (49). Based on this data, the same group further investigated adoptively transferred NKTs in a Phase II clinical trial and showed increased NKT tumor infiltration and significant anti-tumor responses (50). In the United States, a Phase 1 clinical trial by Balk et al. (NCT00631072) recently completed accrual and further demonstrated safety of NKTs. Patients received up to  $2.17 \times 10^8$  NKTs per infusion which was repeated twice with 2-week intervals without significant adverse events (51). In aggregate, the clinical experience with NKTs demonstrates that: i) NKT-cell therapies are well tolerated; ii) the transferred NKTs were rarely persistent iii) objective clinical responses were achieved in some patients. The lack of knowledge of the mechanisms by which NKT

cells exert their anti-tumor activity has previously prevented the appropriate selection of patients for study. In this study, we will focus on patients with recurrent/refractory neuroblastoma, for whom the role of TAMs in the disease outcome is established and who could benefit the most from NKT-cell based immunotherapy. Moreover, by incorporating GD2-CAR expression in the NKT cells we will for the first time provide them with direct anti-tumor targeting, the lack of which has not been fully appreciated in the previous trials.

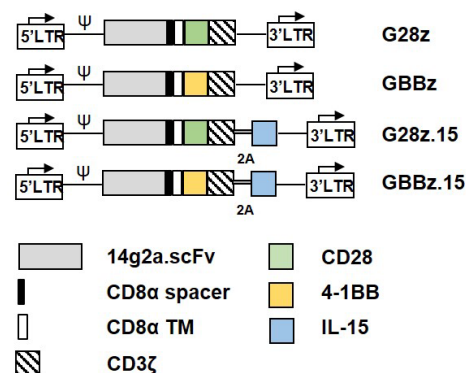
**3.6. Natural Killer T-cells and the role of costimulation in their homeostasis:**

Natural Killer T-cells (NKTs) are an evolutionary conserved sub-lineage of T cells that are characterized by the expression of an invariant T-cell Receptor  $\alpha$ -chain (V $\alpha$ 24-J $\alpha$ 18) and reactivity to self- and microbial-derived glycolipids presented by monomorphic HLA class-I-like molecule CD1d (52). The T-cell receptor/CD1d mediated NKT cell activation by itself is insufficient for expansion and maintaining function of NKTs and additional costimulatory signals are required (53). Two of the most studied costimulatory receptors are CD28 and 4-1BB (CD137), members of the immunoglobulin and TNF/TNFR receptor superfamilies respectively. Human NKTs constitutively express CD28, and blocking antibodies to CD28 or its ligand B7 inhibit  $\alpha$ GalCer-induced NKT proliferation (54-56). 4-1BB mediated costimulation increases the proliferation of iNKTs and their ability to kill macrophages (57) and increases the antitumor effect of iNKTs by increased expansion of tumor specific CD8<sup>+</sup> effector T cells (58). Thus, costimulation by CD28 and 4-1BB is important for the antitumor function of iNKTs.

**3.7. Development of GD2-CAR for Natural Killer T cells:**

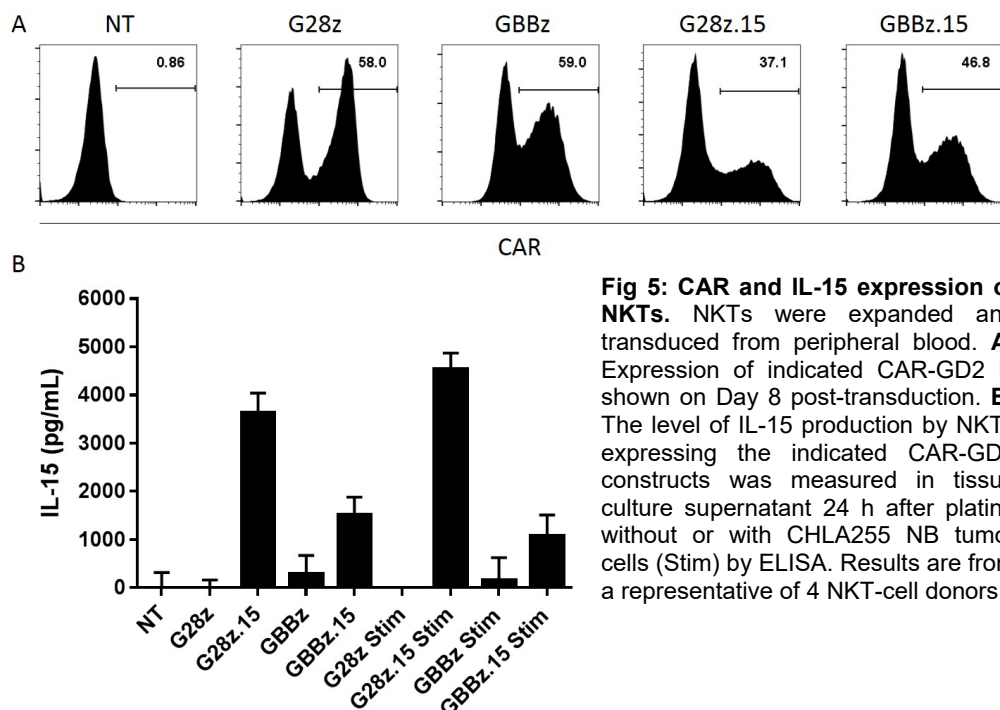
- 3.7.1. Rationale for incorporating costimulatory endodomains in CARs used for NKT cells: When CAR expressing T cells engage the antigen through the CAR, lytic activity may result, but the cells do not receive the cascade of co-stimulatory signals that are required for full activation and proliferation (59-61). We hypothesize that – similarly to T cells – the incorporation of additional co-stimulatory endodomains will improve proliferation of the GD2 specific NKT cell product and increase resistance to an immunoinhibitory tumor environment, thereby increasing the frequency, persistence and anti-tumor activity of the GD2-CAR NKT cells. Normally, full activation of T cells and NKTs is accomplished after they bind to a target via their native receptor and receive stimulation from activating ligands on antigen presenting cells (APCs) as well as cytokines secreted in the local microenvironment (53, 62). Most tumor cells lack the additional costimulatory ligands needed for enhanced NKT cell activation, and 2<sup>nd</sup>-generation CARs have therefore been engineered to include a co-stimulatory domain.

3.7.2. Rationale for incorporating Interleukin-15 (IL-15) in the constructs: Co-expression of cytokines in tumor specific effectors is an attractive strategy to manipulate the tumor microenvironment and/or to support the survival of the adoptively transferred lymphocytes. A recent Phase 1 study tested this concept by evaluating the safety and antitumor efficacy of tumor infiltrating lymphocytes engineered to express IL-12 (63) and an ongoing study is testing T cells co-expressing IL-12 with a 2<sup>nd</sup>-generation MUC16ecto-specific CAR (NCT02498912). In this study, we provide a rationale for inclusion of another cytokine, IL-15, in the CAR design. IL-15 is critical during development and peripheral maintenance of CD8 T, NK, and NKT lymphocytes (64). Preclinical testing of CAR T cells co-expressing IL-15 showed improved antitumor efficacy (65, 66). IL-15 is a key cytokine for iNKT cell homeostasis and antitumor activity. Our lab found that IL-15 protected antigen-activated iNKT cells from hypoxia which otherwise can inhibit their function in the tumor microenvironment. Moreover, transgenic expression of IL-15 in adoptively transferred iNKT cells dramatically enhanced their antimetastatic activity in mice (43). Thus, to overcome the inhibition of tumor microenvironment and enhance iNKTs survival after adoptive transfer, combining IL-15 with the expression of GD2-CAR may enhance the therapeutic activity of iNKTs against cancer. GD2-CAR constructs: Using the clinically validated 14g2a scFv, the following four constructs were generated: 14G.28 (G28z), 14G.28-IL-15 (G28z.15), 14G.BB (GBBz), and 14G.BB-IL-15 (GBBz.15) as shown in **Fig 4**. The cDNA sequence of IL-15 was added to G28z and GBBz following a 2A sequence, allowing two separate peptide fragments to be produced from a single messenger RNA.

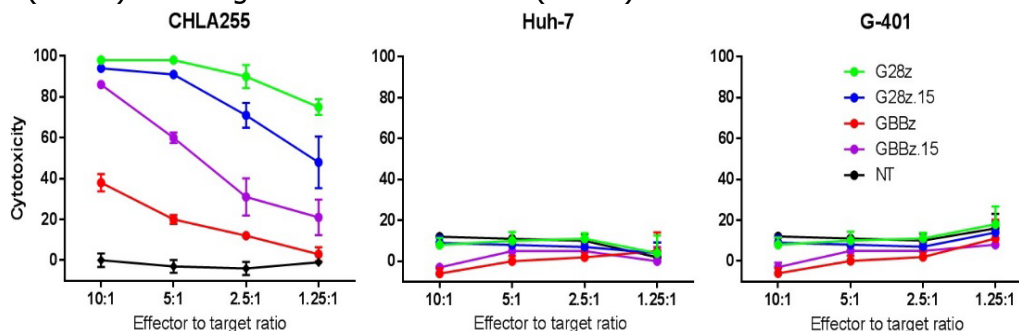


**Figure 4: Schematic map of GD2-CARs.**  
TM: transmembrane.

**3.7.3. Newly generated retroviral constructs can stably express the GD2-CAR and IL-15 in natural killer T (NKT) cells:** NKTs were transduced with the constructs above using a retroviral vector. All constructs were stably expressed by NKTs with transduction rate ranging from 37.1% to 59.0% as determined by flow-cytometry (**Fig 5A**). In addition, IL-15 was expressed and secreted by NKTs transduced with IL-15 containing constructs as measured by ELISA analysis of tissue culture supernatant 24 h after plating of NKTs. Lastly, IL-15 production is maintained following activation of NKTs through CAR stimulation by GD2+ CHLA255 NB cells (**Fig 5B**).



**3.7.4. Natural Killer T cells expressing GD2-CARs specifically kill GD2+ tumor cell:** To determine the specificity and cytotoxicity of GD2-CAR NKTs, firefly luciferase expressing GD2+ NB (CHLA255) and GD2- hepatocellular carcinoma (Huh-7) or malignant rhabdoid tumor (G-401) cells were co-cultured with GD2-

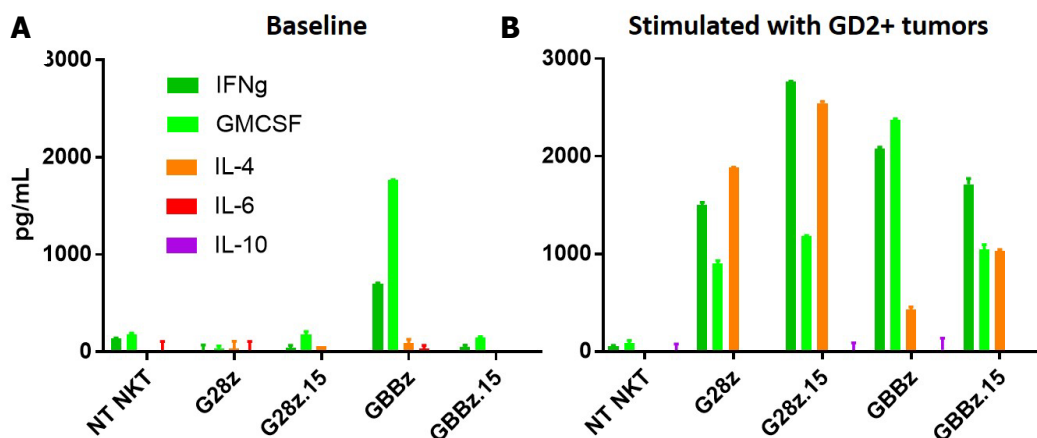


**Fig 6. Cytotoxicity of GD2-CAR NKTs.** NKTs were transduced with indicated constructs and co-cultured for 4 h with Ffluc expressing GD2+ (CHLA255) and GD2- (Huh7 and G-401) tumor cells at various effector-to-target ratios. Tumor cell Ffluc expression was measured at 4 h to calculate the degree of tumor cell lysis. Results are from a representative of four NKT-cell donors.

CAR NKTs. Bioluminescence was measured after 4 h and was used to calculate

specific tumor cell lysis. GD2-CAR NKTs specifically and effectively kill GD2+ NB tumor cells and do not kill GD2- tumor cells (**Fig 6**).

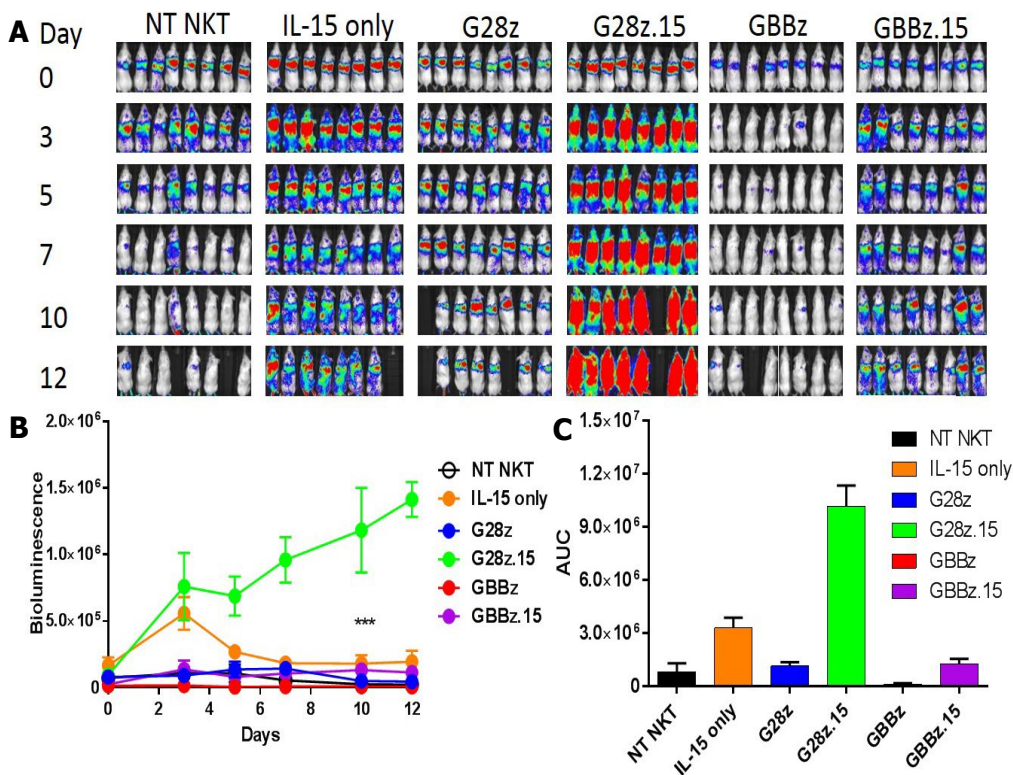
**3.7.5. GD2-CAR NKTs produce cytokines in an antigen-dependent manner.** Upon activation NKTs can produce Th1 (GM-CSF and IFN- $\gamma$ ) and Th2 (IL4 and IL-10) cytokines. If CARs form clusters on the cell surface in the absence of cognate antigen, the triggered tonic signaling can lead to cell exhaustion which in turn results in decreased antitumor effect (67). Tonic signaling can be detected by spontaneous cytokine release by resting cells. Thus, we measured the indicated cytokines in tissue culture supernatants of CAR NKTs and non-transduced controls



**Fig 7. Cytokine production of GD2-CAR NKTs.** NKTs were transduced with indicated constructs and co-cultured with GD2+ (CHLA255) tumor cells at 1:1 effector to target ratio. Cytokine levels were measured in tissue culture supernatant at 24 h with Luminex multiplex bead assay according to the manufacturer's instructions. Results are from a representative of 4 NKT-cell donors.

with or without stimulation. We found that GBBz construct induced significant IFN $\gamma$  and GM-CSF production in the absence of antigen, but co-expression of IL-15 in the same transgene reversed this effect (**Fig 7A**). Our lab has shown that the CAR induced cytokine release pattern of CAR NKTs can be manipulated by the composition of the CAR (68). To determine the antigen-specific cytokine production of CAR NKTs, we co-cultured GD2+ CHLA255 NB cells with CAR NKTs and non-transduced NKTs at 1:1 ratio and cytokine levels in tissue culture supernatants were measured at 24 h. (**Fig 7B**). All CAR constructs induced cytokine production in NKTs stimulated with GD2+ tumor cells. We confirmed our previous finding that 4-1BB induces more IFN- $\gamma$  and GM-CSF, while CD28 induces more IL-4 production. NKTs did not produce IL-10 which has immunomodulatory properties and can decrease the effector function of CD8+ T cells or IL-6, a cytokine likely a key mediator of toxicities detected after CAR T cell therapies (69).

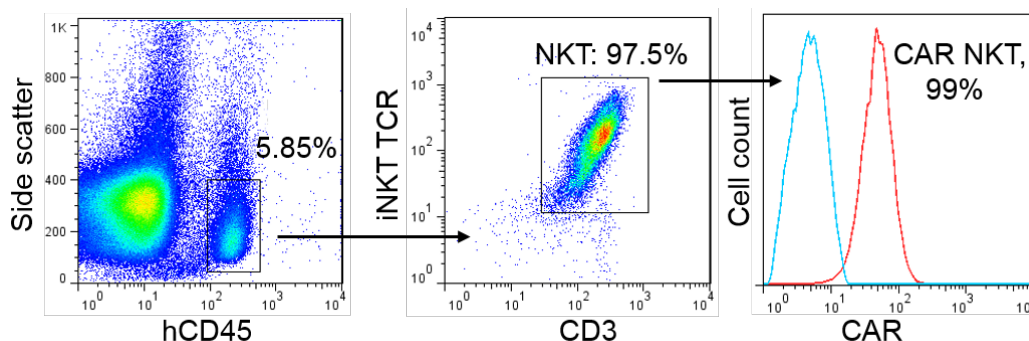
**3.7.6. G28z.15 construct supports superior NKT cell in vivo persistence.** Antigen-dependent expansion and long-term persistence of CAR expressing cells is a key aspect of effective adoptive cell therapies (70-73). We examined the in vivo persistence of NKTs after transfer to NSG mice with 3-week established NB xenografts (**Fig 8**). In this setting, NB cells were unlabeled while NKTs were transduced with a high-intensity firefly luciferase cDNA (Ffluc). NKTs expressing Ffluc only or Ffluc plus IL-15 served as controls. The transgenic expression of IL-15 improved the persistence of NKTs compared NT-NKTs and, NKTs expressing



**Fig 8: Expansion of NKTs expressing GD2-CARs:** NSG mice were injected i.v. with  $2 \times 10^6$  CHLA255 NB cells followed on Day 21 by i.v. injection of  $10^7$  Ffluc+ NKTs. **A:** Serial bioluminescence imaging showing NKT cell expansion. **B:** Quantified bioluminescence for each treatment group at indicated time points **C:** Bioluminescence over time derived area under the curve (AUC) of each treatment groups.

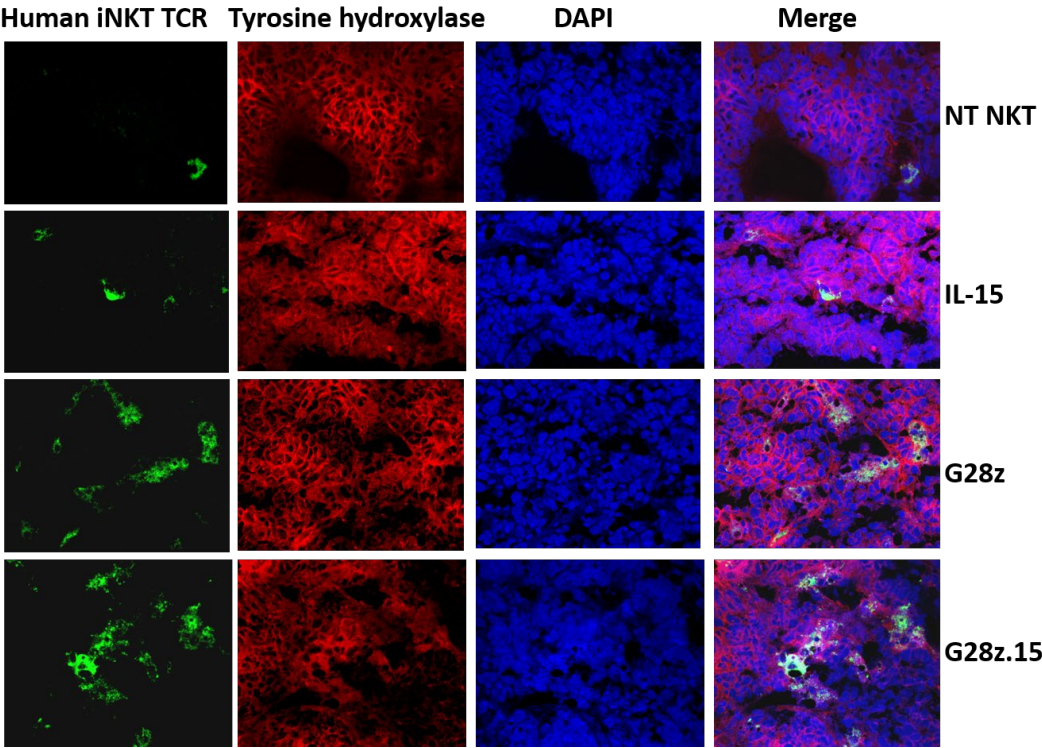
IL-15 only did not expand in the absence of CAR stimulation. For the GD2-CAR groups, NKTs transduced with the GBBz construct were largely eliminated three days after injection while NKTs transduced with the G28z construct were readily detectable 12 days after injection. Compared to 4-1BB alone (GBBz), co-expression of 4-1BB and IL-15 (GBBz.15) in the CAR construct modestly improved NKT cell persistence. However, co-expression of IL-15 with CD28 in the CAR (G28z.15) had a much more dramatic effect on NKT cell persistence compared with the corresponding construct with CD28 alone (G28z). In fact, G28z.15 CAR NKTs underwent substantial expansion within the first 12 days. Consistent with day 12 whole-animal quantitative bioluminescent analyses (**Fig 8 B & C**), flow cytometry analysis of spleen, liver, bone marrow, and lung tissues after animal sacrifice on day 13 confirmed that mice in the G28z.15 group had the highest frequencies of human NKTs compared with other groups (data not shown). In a separate experiment, we analyzed GD2-CAR NKT persistence over an extended

time course and confirmed that only the G28z.15 construct enabled long-term NKT cell persistence in numerous tissue sites. For example, the frequency of G28z.15 CAR NKTs reached 5.8% of bone marrow cells 30 days after adoptive transfer (**Fig 9.**). Similarly, we observed long-term persistence of NKTs transduced with G28z.15, but not other constructs, in spleen, liver, and lung tissues—common sites of neuroblastoma metastases in this model (data not shown).

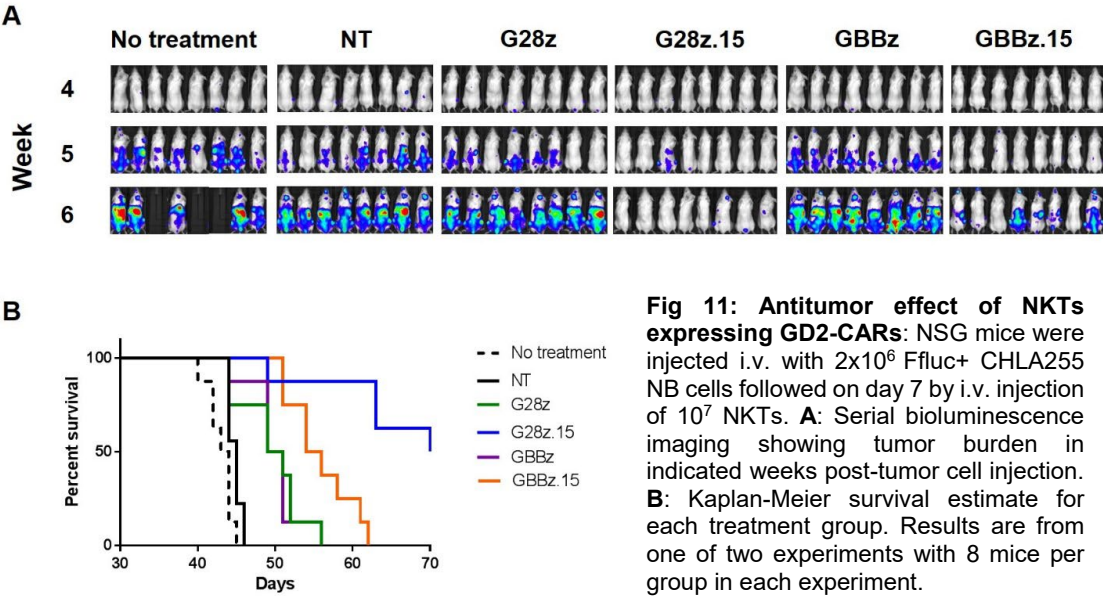


**Fig 9: Long-term persistence of G28z.15 NKTs in bone marrow:** NSG mice were injected i.v. with  $2 \times 10^6$  CHLA255 NB cells followed 16 days later by i.v. injection of  $10^7$  Ffluc+ CAR NKTs. Mice were sacrificed at different time points after CAR NKT cell transfer. Bone marrow was derived from femurs and analyzed by flow cytometry for the frequency of human CAR NKT cells using the indicated gating strategy. CAR expression was detected with 1A7 anti-idiotypic mAb (red) using mouse IgG as a negative control (blue). Shown are representative FACS plots from the bone marrow of one of eight mice in the G28z.15 group 49 days post-CHLA255 NB cell injection/33 days post-NKT cell injection.

**3.7.7. G28z.15 construct mediates effective infiltration of NKT cells into tumor tissues.** Localization of therapeutic effector cells to the tumor site is a critical requirement for effective cancer immunotherapy. Therefore, we performed confocal imaging of sectioned tumor tissues from NB tumor-bearing NSG mice that received adoptive transfer of GD2-CAR NKTs as described in **Fig 8**. We analyzed NKT cell infiltration into the liver, a common site of NB metastasis in this model, by performing the following stains on sectioned liver tissue: 1) human NKT cell invariant TCR $\alpha$  chain (6B11 mAb), 2) neuroblastoma cells (tyrosine hydroxylase), and 3) general cell nucleus staining (DAPI). We found that 13 days after adoptive transfer to tumor-bearing NSG mice, there was a higher frequency of NKTs expressing the G28z.15 CAR than the G28z CAR in liver metastases (**Fig 10**). Importantly, GD2-CAR NKTs not only homed to tumor sites and co-localized with tumor cells, but also penetrated deep into the tumor parenchyma.



**Fig 10: GD2-CAR NKT cells penetrate into tumor parenchyma:** NSG mice were injected i.v. with  $2 \times 10^6$  CHLA255 NB cells followed 27 days later by i.v. injection of  $10^7$  Ffluc+ CAR NKTs. On day 13 after CAR NKT cell transfer, mice were sacrificed and NKT tissue infiltration into liver tissue sections was analyzed by confocal microscopy. Shown are NKT cells (green) in tumor tissues (red) in the liver (all cells – blue). The images show increased tumor infiltration of NKTs transduced with G28z.15 compared with G28z, IL-15, or non-transduced (NT) NKTs. Results are from a



**Fig 11: Antitumor effect of NKTs expressing GD2-CARs:** NSG mice were injected i.v. with  $2 \times 10^6$  Ffluc+ CHLA255 NB cells followed on day 7 by i.v. injection of  $10^7$  NKTs. **A:** Serial bioluminescence imaging showing tumor burden in indicated weeks post-tumor cell injection. **B:** Kaplan-Meier survival estimate for each treatment group. Results are from one of two experiments with 8 mice per group in each experiment.

**3.7.8. G28z.15 construct enables superior therapeutic activity of GD2-CAR NKTs *in vivo*.** To compare the *in vivo* therapeutic potential of GD2-CAR constructs, we injected GD2-CAR NKTs into a metastatic NB mouse model (NSG mice). NSG mice were intravenously injected with CHLA255 NB cells stably expressing Ffluc. Seven days later, mice were infused intravenously with NKTs transduced with the constructs shown in **Fig 4**. Mice received no treatment or non-transduced (NT) NKTs were used as negative controls. **Fig 11** demonstrates that NKTs expressing G28z.15 had significantly more anti-tumor activity against NB xenografts than NKTs expressing any other CAR construct or NT NKTs ( $P < 0.001$ ). Indeed, a single injection of G28z.15 NKTs resulted in a median survival of 70 days post-tumor cell injection compared to a range of 42 – 53 days in other groups. In a repeat therapeutic experiment, we used a more aggressive tumor model in which we delayed treatment initiation with CAR NKTs from day 7 to day 16 after tumor injection. Under these conditions, the 4-1BB-containing construct (GBBz) failed to provide a survival benefit unless expressed in combination with IL-15. In contrast, both CD28-containing constructs (G28z and G28z.15) had potent anti-tumor activity that significantly prolonged animal survival ( $P < 0.001$ , data not shown). Although G28z.15 did not prolong survival compared with G28z or GBBz.15 groups, pathological analysis at the time of sacrifice revealed that only mice treated with G28z.15 CAR NKTs had no detectable tumor cells in the liver and adrenal glands, common sites of NB metastasis in this model and in NB patients. Thus, consistent with the unique ability of the G28z.15 construct to induce *in vivo* expansion, superior tumor infiltration, and long-term persistence of NKTs, G28z.15 NKTs demonstrated the most potent anti-tumor activity in a xenogenic mouse model of neuroblastoma.

### **3.7.9. Safety**

#### **Evaluation of toxic side-effects of G28z.15 NKT cell treatment in mice**

In order to evaluate potential toxic side-effects of G28z.15 NKT cells, NSG mice with 12-day established metastatic neuroblastoma xenografts received adoptive transfer of  $10^7$  Ffluc+ G28z.15 NKT cells as described in **Fig 8**. Mice were monitored for NKT cell persistence and expansion by bioluminescence imaging every 2-3 days. In parallel, clinical symptoms were monitored daily and mouse weight was measured every other day. Similar to results shown in **Fig 8**, G28z.15 NKT cells expanded in the mice within the first two weeks and then remained at similar levels for the following two weeks as measured by bioluminescence. At day 12 post-NKT cell injection, we observed mild clinical symptoms manifested in hunched stature and ruffled fur. On that same day, half of the mice received methyl-prednisolone (MPD, 0.43 mg/mouse, calculated to be equivalent to the human therapeutic dose of 1.4 mg/kg), while the remaining half was left untreated. The observed clinical symptoms resolved fully within two days in both groups, and none of the eight monitored mice experienced weight loss regardless of MPD treatment. Serum taken on days 12 and 15 was analyzed by ELISA for the presence of mouse (m)IL-6 (a known mediator of toxicity in CAR T cell clinical trials) and human (h)IL-15. There was no detectable mIL-6 or hIL-15 in the serum. In addition, four mice were sacrificed on day 12 just prior to MPD treatment (group 1) and eight mice were sacrificed on day 15, four of which were not treated with MPD (group 2) and four of which were treated with MPD (group 3). A highly experienced clinical pathologist performed detailed pathological examination of tissues from all twelve mice including lung, heart, liver, spleen, kidney, adrenals,

stomach, intestine, pancreas, spine, brain (cerebrum and cerebellum), skin, bone with bone marrow, and skeletal muscle. No features of graft versus host disease were found in any of the mice, and only minimally detectable tumor cells were found in two of 12 mice (from group 2), additional evidence of the potent *in vivo* anti-tumor activity of G28z.15 NKT cells. In groups 1 and 3, no pathologic abnormalities were found in any organ/tissue except for the lungs, where mice in both groups exhibited signs of focal minimal-to-mild interstitial chronic inflammation. These findings are consistent with the mild clinical symptoms of hunched stature and ruffled fur described above. Although these clinical symptoms self-resolved regardless of MPD treatment, the treatment was highly effective in eliminating the residual inflammatory foci in the lungs. In a separate experiment, we evaluated the long-term toxicity of G28z.15 NKT cells in a similar manner and found no evidence of toxicity to normal tissues upon pathological examination of mice 2 – 3 months after transfer of G28z.15 NKT cells.

**Absence of autonomous growth of GD2-CAR NKTs expressing IL-15.**

Leukemic transformation has been observed in IL-15 transgenic mice (74, 75). In those studies, however, IL-15 was expressed in all cells of experimental animals including stem cells, which are known to be susceptible to oncogenic transformation. Transduction of human CD8 T cells with IL-15 resulted in the outgrowth of one abnormal clone out of 26 cultures (76). However, as the authors acknowledged, even that clone did not meet the criteria for malignant transformation because, “it could not be grown under limiting dilution conditions, in soft agar, nor could the cell line be established in immunocompromised mice.”

Our results demonstrate that even after two and half months of persistence in NSG mice, G28z.15 NKTs lacked the ability to proliferate in the absence of TCR stimulation and did not survive in the absence of exogenous IL-2 (see section 8.2.5). Therefore, IL-15-mediated oncogenic transformation has not been documented in human cells and is not supported by the results of our experiments with G28z.15 NKTs.

**IL-15 is not detectable in mice after infusion with G28z.15 NKTs.**

Systemic administration of hIL-15 in patients with cancer has been associated with dose-limiting toxicities including hypotension, thrombocytopenia, and elevated liver enzymes (77). To evaluate hIL-15 levels in tumor-bearing mice after adoptive transfer of  $1 \times 10^7$  G28z.15 NKTs, blood samples were taken at peak CAR NKT expansion. These samples were tested by ELISA and no detectable levels of hIL-15 were found. Since the dose given to each mouse is equivalent to  $1.5 \times 10^9/m^2$  in humans, we do not expect systemic hIL-15-related toxicities with G28z.15 NKTs.

**Summary of preclinical testing:** NKTs expressing G28z.15 showed specific killing of GD2+ targets; antigen-dependent effector cytokine production; superior *in vivo* expansion and persistence; and potent, well-tolerated therapeutic activity in a xenogenic metastatic model of neuroblastoma in NSG mice. **Thus, the presented data justify further clinical development and support our plan to evaluate the safety, define the maximum tolerated dose, and measure the anti-tumor activity of G28z.15 NKTs in a preliminary fashion.**

## 4. Study Rationale

### 4.1. **Rationale for selecting Neuroblastoma as a target:**

Neuroblastoma is a form of pediatric cancer that leads to significant morbidity and mortality. As the most common extracranial malignant solid tumor of childhood, it accounts for greater than 10% of all pediatric cancer deaths. With current therapy, overall survival remains less than 40% for patients with stage 4 high-risk disease and there is an urgent need for improved treatment options for these children. We and other groups have shown immunological and clinical efficacy with the use of neuroblastoma-specific immunotherapy targeting GD2. However, while the use of GD2 specific CAR expressing activated T cells was associated with clinical responses, especially for those in whom we were able to detect cells for more than 6 weeks after infusion, detected cells were present at very low frequencies within the peripheral blood.

Applying CAR-redirectioned immunotherapy to the treatment of solid tumors, including neuroblastoma represents a critical challenge because immunosuppressive tumor microenvironment impairs T cell localization, persistence, and the execution of their effector function. Therefore, improving homing and survival of CAR-redirectioned effector cells at the tumor site is of paramount importance for achieving clinical efficacy of this therapeutic modality in solid tumors. We hypothesize that NKT cells will have a major advantage as a CAR carrier because, unlike conventional T cells, NKT cells have natural tumor homing abilities and NKT-cell infiltration of primary tumors is associated with good outcome in children with stage 4 neuroblastoma (36, 78). NKTs cannot directly target neuroblasts and therefore without additional modification, cannot significantly control tumor growth. With the incorporation of GD2-CAR, this limitation is overcome, and data from in vitro and mouse experiments suggests that such NKTs will be able to destroy neuroblasts via GD2-CAR and kill TAMs by their innate TCR. Moreover, NKT-mediated killing or inhibiting TAMs may be critical, as they are a key element of tumor-promoting inflammation which is an independent prognostic factor of extremely poor outcome in MYCN non-amplified neuroblastoma (79). Our preclinical studies demonstrate that GD2-CAR NKT cell function and anti-tumor activity can be further enhanced using a co-expression of a 2<sup>nd</sup>-generation GD2-CAR with CD28 costimulatory endodomain and the homeostatic cytokine IL-15 (G28z.15). This combination provides optimal balance between NKT cell survival and Th1-like functional polarization resulting in improved in vivo persistence and anti-tumor activity without evidence of severe toxicity.

### 4.2. **Rationale for initial dose level:**

Multiple Phase 1 clinical trials evaluated the safety of non-transduced NKTs outside the United States and these trials demonstrated that NKTs are well tolerated (49, 50). A Phase 1 clinical trial by Exley et al. (NCT00631072) recently completed accrual and further demonstrated safety of NKTs. In this trial, patients received up to  $2.17 \times 10^8$  NKTs per infusion. Each patient received three infusions at 2-week intervals without significant adverse events (51). Based on these data, we will use the initial dose level of  $3 \times 10^6/m^2$  G28z.15 NKTs. CAR NKT cell dose will be calculated based on the CAR+ cells. The lowest acceptable CAR transduction criterium will be set at 20% or greater for product release by QC. The initial dose of CAR NKTs on dose level 1 will be the

$3 \times 10^6/\text{m}^2$  of CAR+ cells. In case of 20% transduction efficiency, the total number of infused cells will be  $1.5 \times 10^7/\text{m}^2$  cells. This initial dose is 8-fold less than the cell dose already given safely by Exley et al (51). Similar to our other CAR T cell studies, we will increase the dose gradually in 3-fold increments (IND 14036[HERCREEM], IND 14403 [HERT-GBM], and IND 14050 [iCAR]). Targeting GD2 with CAR T cells has also been evaluated in patients with relapsed or refractory neuroblastoma and doses up to  $2 \times 10^8/\text{m}^2$  of CAR T cells have been safe and well tolerated (IND 13149 [NESTLES] and IND 15609 [GRAIN]) (33, 34). Thus, we expect that on DL1 of a single dose of G28z.15 NKTs cells, the probability of toxicity will be minimal in our study.

Those patients who have no evidence of disease progression at their evaluation 4 weeks post-infusion and who did not experience a dose-limiting toxicity during this time, may be eligible to receive additional doses of G28z.15 NKTs at their original cell dose with lymphodepletion. Patients will need to meet treatment eligibility at the time of each start of lymphodepletion for reinfusions.

#### 4.3. **Rationale for Lymphodepletion**

Lymphodepletion has been shown to improve the anti-tumor efficacy of adoptive cell therapies by direct antineoplastic effect of the used chemotherapeutic agents and by improving survival and function of adoptively transferred cells (80). The size of the lymphoid compartment is tightly controlled by homeostatic factors, including the access to antigen-presenting cells (APCs), major histocompatibility complex (MHC) presenting self and antigenic peptides, and direct influence of host T lymphocytes including regulatory cells CD4+CD25+FoxP3+ Treg, as well as the competition for cytokines such as IL-2, IL-7, IL-15 and IL-21 (81-89). Administration of certain lymphodepleting chemotherapy or total body irradiation, reduces the patient's lymphoid cell pool, improves the adoptively transferred cells access to APCs, peptides presented by MHCs and homeostatic cytokines resulting in superior expansion and persistence of transferred cells (80, 90, 91). Most importantly, after lymphodepletion, long-term complete remissions are achievable in heavily pretreated cancer patients with genetically engineered adoptive cell therapies (92-94). The most commonly used lymphodepleting regimens combine cyclophosphamide and fludarabine (cy/flu) and a fractionated cy/flu regimen was used in the only clinical protocol to date resulting in sustained complete remission in a patient with bulky disease (80, 94, 95). Therefore, we will use a fractionated cy/flu regimen for lymphodepletion in this trial.

#### 4.4. **Rationale for encoding IL-15 cDNA within GD2-CAR construct.**

It is widely accepted that microenvironment of solid tumors is highly immunosuppressive and not permissive for therapeutic effector cells including NKTs or CAR NKTs. A rational way to overcome this barrier is to consider natural homeostatic requirements of the effector cells. IL-15 serves as a homeostatic cytokine responsible for maintenance of NKTs in peripheral tissues physiologically and we have demonstrated that transgenic expression of IL-15 in NKTs restores their functionality and antitumor activity in a humanized model of neuroblastoma (43). Our preliminary data in support of this application further demonstrates that IL-15 expression within GD2-CAR enables long-term in vivo persistence and superior antitumor activity of CAR NKTs in a metastatic neuroblastoma model. These results are consistent with the ability of systemically administrated human recombinant IL-15 to activate NK and

memory CD8 T cells and mediate anti-tumor responses in a phase I clinical trial (77). However, with the systemic administration of IL-15 in that trial high, up to 1260 pg/mL serum IL-15 levels were reached and dose-limiting toxicities were observed. By contrast, IL-15 was not detectable systemically in mice treated with G28z.15 NKTs even at the peak of NKT-cell expansion. This is consistent with the tissue localization of NKTs and local consumption of the produced IL-15. Furthermore, superior therapeutic activity of G28z.15 NKTs was not associated with severe toxicity in mice.

## **5. Risks of this Study**

Potential toxicities may be categorized as those related to: i) infusion of NKT cells; ii) general consequences of retroviral transduction; iii) specific consequences of the transgene expressed (targeting GD2, cytokine release syndrome and neurotoxicity, IL-15 expression); and iv) risks of using fludarabine and cyclophosphamide.

### **5.1. Infusion of CAR NKT cells**

In this study, we will be administering transduced, autologous, peripheral blood-derived, activated NKT cells, a subset of T cells modified to express a CAR directed against GD2, and to express and secrete human IL-15. Because of the limitations involved in purifying specific cell subsets from patient-derived starting materials, the infusion product will contain some percentage of non-transduced NKT and other T cells. Many previous studies have infused larger numbers of ex-vivo cultured NKT cells with no adverse effects (50, 51). Thus the risk from infusing autologous NKT cells will be minimal.

### **5.2. General Consequences of Retroviral Transduction**

Retroviral transduction results in new random integrations in host cell DNA, which rarely may cause abnormal or uncontrolled proliferation (52). This effect is much more common with replication-competent retrovirus (RCR), where each cell receives multiple integrants (53). We will test the producer line and all batches of supernatant with biological assays to confirm the absence of detectable RCR to exclude this possibility.

There has been concern that even a single retroviral integration can contribute to oncogenesis when hematopoietic stem cells are transduced. Fischer's group replaced the missing common gamma chain ( $\gamma_c$ ) in X-linked severe combined immunodeficiency (X-SCID) patients with ex-vivo retroviral transduction of autologous stem cells. Four of 11 treated patients developed T-cell acute lymphoblastic leukemia (ALL) associated with a single retroviral integrant (54-56). More recently 1 of 12 patients in a similar study in England has also developed T-cell lymphoproliferation (57). In 4 of 5 patients the integration site of the retrovirus was similar.(58) Malignancies have also been reported in 2 of 6 patients after gene therapy for X-linked chronic granulomatous disease (X-CGD) and in 1 of 10 for Wiskott-Aldrich syndrome (WAS).(58-60) As concerning these findings are for the prospect for gene therapy of X-SCID, X-CGD, and WAS it cannot be generalized to the entire field of gene therapy, especially since none of the patients with adenosine deaminase (ADA)-SCID developed malignancies after gene transfer after long-term follow up (61).

It is likely that correction of  $\gamma_c$  deficiency and related immunodeficiency syndromes represent special cases. For example, the  $\gamma_c$  is a shared component of interleukin (IL)-2, IL-4, IL-7, IL-9 and IL-15. Hence it is a crucial component T-cell proliferation and thymogenesis. A proliferative advantage is expected for progeny of stem cells with functional  $\gamma_c$ . In female carriers of X-SCID there is a pattern of non-random X-inactivation in T-cells, B-cells and NK-cells (62). Moreover, a patient with X-SCID developed substantial numbers of T-cells following reversion of the mutant allele in a single hemopoietic stem cell (63). High efficiency retroviral transduction of human stem cells is difficult to achieve even with GALV pseudotyping. It is likely that in X-SCID few truly pluripotent stem cells were transduced and the stem cell pool expressing the highest level of transgene (due to integration at a transcriptionally active site) then undergo numerous doublings to restore the entire T-cell compartment. Random mutations caused by this supra-physiological proliferation combined with retroviral integration in a transcriptionally active region led to leukemogenesis.

Patients eligible for the current study will have relapsed or refractory neuroblastoma. In light of this, the natural history and poor prognosis of this disease, and given the entire previous experience with retroviral gene therapy, we feel that the small risks of retrovirally-induced leukemogenesis are small and are justified in these patients.

### 5.3. **Specific consequences of transgenes expressed**

#### 5.3.1. Toxicity from targeting GD2 antigen with CAR NKT cells

Although GD2 is expressed on peripheral nerves, we did not detect DLTs related to peripheral nerve injury or irritation in our already published clinical trial of CAR T cells (33, 34, 96, 97) or among patients who are treated on our most recent clinical trial (IND 15609 [NCT01822652]). The most likely peripheral nerve injury/irritation related symptoms are pain and neurotoxicity: Pain will be assessed at each visit or as needed by objective pain score system or FACES scale for children who cannot quantify their pain. Neurologic examination will evaluate any potential other neurologic symptoms at each visit or more often if needed based on symptomatology. Should G28z.15 NKTs produce direct toxicity to GD2-expressing normal tissues, a number of therapeutic maneuvers are available. These include corticosteroids in high doses or anti-T-cell antibodies such as Campath 1H, both of which will deplete the majority of circulating transduced cells.

#### 5.3.2. Cytokine Release Syndrome and Neurotoxicity

Cytokine release syndrome (CRS) and neurotoxicity is a known side effect in approximately 30% or more patients who received T cells (98) or bispecific T-cell engager molecules (99, 100). The majority of CRS have been reported after the infusion of CAR T cells (101), but CRS can also occur after the infusion of conventional antigen-specific T cells (102) or tumor infiltrating lymphocytes (103). Patients will be monitored closely as per study calendar and assessed for evidence of incipient CRS (onset of fever, malaise and dyspnea) and treated promptly. Management of CRS will follow published guidelines (69, 98), and is described in more detail in SOP F 05.11.XX. There have been several reports of neurotoxicity (e.g. encephalopathy, somnolence, aphasia) after the infusion of

CD19-specific CAR T cells (98). Patients will be monitored closely as per study calendar and assessed for evidence of incipient neurotoxicity (mild somnolence, drowsiness or sleepiness) and treated promptly. Management will include IL-6 receptor antibody (4-8 mg/kg), and/or steroids (1-2 mg/kg/day of methylprednisolone or equivalent), and prophylactic anti-epileptics as described in the literature and our SOP F 05.11.XX (98).

5.3.3. IL-15 production by NKT cells:

The systemic administration of IL-15 has been evaluated in a Phase 1 study showing promising antitumor properties and acceptable safety profile at the lowest dose level (0.3 µg/kg/day for 12 doses); however, higher systemic doses were associated with dose-limiting toxicities of hypotension, thrombocytopenia and elevation of AST/ALT enzymes (77). CD19-CAR T cells engineered to express IL-15 have been tested in preclinical models to evaluate the effect of locally produced IL-15. These studies showed improved antitumor efficacy compared to T cells expressing CD19-CAR alone without reported toxicity in mice (65, 66). Our own preclinical results demonstrate the beneficial effect of IL-15 on the antitumor properties of GD2-CAR NKTs. Our safety assessment showed no toxicity to murine tissues by histologic evaluation, undetectable systemic levels of IL-15 after adoptive transfer of G28z.15 NKTs and a lack of autonomous growth of G28z.15 NKTs. Although these data support a low risk of additional toxicity from the incorporated IL-15, the co-expression of IL-15 in tumor-redirectioned NKTs has not been tested in humans. Since patients eligible for the current study will have relapsed or refractory neuroblastoma, and given the apparent safety in preclinical testing, we believe that the potential benefits of co-expressing IL-15 with a GD2-CAR in NKTs outweigh the potential risks.

A Grade 5 adverse event occurred 17 days after the first subject was treated with G28z.15 NKT cells at dose level 5 ( $3 \times 10^8$  cells/m<sup>2</sup>). This subject was determined to have human metapneumovirus (HMPV) infection (by PCR) and Grade 2 CRS that was treated with immunomodulators on Week 1. The following week, the subject developed polyclonal hyperleukocytosis with peripheral WBC exceeding 200,000/uL complicated by multiorgan dysfunction. Several cytokine and chemokine serum concentration including IL-15 had increased a few days before the subject's death. Based on our root cause analysis, this SEA was probably due to change in G28z.15 NKTs manufacturing resulting in the generation of a product with high proliferative capacity. The manufacturing process was changed back to the original method which had been generating safe products for 13 treated patients prior to this event. We also modified our releasing criteria with a modified autonomous growth assay. Monitoring and management of hyperleukocytosis (three times the upper limit of normal value for age) will be performed as described in Section 8.4.8.

5.4. Lymphodepletion-related toxicities

In this protocol, we will use fractionated 500 mg/m<sup>2</sup>/dose x2 cyclophosphamide and 30 mg/m<sup>2</sup>/dose x3 fludarabine for lymphodepletion (Section 7.2). This has been a widely used lymphodepletion protocol prior to CAR T cell infusion (36, 46, 71). Both

cyclophosphamide and fludarabine are commercially available and a list of their toxicities is attached in **Appendix B**.

## 6. Study Enrollment and Patient Eligibility

- 6.1. **Informed Consent/Assent:** The objectives, procedures and treatments in this trial, including risks, benefits and discomforts, and potential alternative therapies will be carefully explained to the patient or the patient's parents/guardian if the patient is a child. Signed informed consent and assent will be obtained according to institutional guidelines.
- 6.2. **Patient Registration:** Prior to study enrollment, all patients must have been registered with a Cell and Gene Therapy research coordinator.
- 6.3. **Institutional Pathology Report:** At the time of enrollment, the institutional pathology report for the diagnosis of high-risk neuroblastoma must be faxed to the Center for Cell and Gene Therapy research coordinator at (832) 825-9622.
- 6.4. **Study Enrollment:** Patients may be enrolled once all eligibility requirements for the study have been met.
  - 6.4.1. Minimum waiting period after first patient on each dose level: Recent clinical studies report that CAR T cell infusion related toxicities are typically expected to occur within the first 14 days post-infusion (70, 71, 73, 104, 105); however, a Grade 5 SAE occurred on this study past Day 14, thus, 28 day minimum waiting period will be implemented before proceeding to enroll the second patient on each dose level.
  - 6.4.2. Minimum waiting period to enroll subsequent patients: Subsequent patients after the 2<sup>nd</sup> patient may be enrolled every week on each dose level.
- 6.5. **Patient Criteria:** All clinical and laboratory data required for determining eligibility must be available in the patient's medical/research record which will serve as the source document.
  - 6.5.1. Procurement Inclusion Criteria
    - 6.5.1.1. Relapsed or refractory high-risk neuroblastoma
    - 6.5.1.2. Life expectancy of at least 12 weeks
    - 6.5.1.3. Age greater than 1 year and less than 21 years old
    - 6.5.1.4. Karnofsky/Lansky score of 60% or greater (**Appendix A**)
    - 6.5.1.5. Absence of HAMA prior to enrollment (only in patients that have been previously treated with murine antibodies)
    - 6.5.1.6. Ability to tolerate leukocyte apheresis

- 6.5.1.7. Informed consent and assent (as applicable) obtained from parent/guardian and child.
  - 6.5.1.8. Patients must have an ANC  $\geq 500/\mu\text{l}^{\#}$ , platelet count  $\geq 20,000/\mu\text{l}$ . Patients may be transfused to obtain a platelet count  $\geq 20,000/\mu\text{l}$ .
  - 6.5.1.9. Pulse Ox  $\geq 90\%$  on room air
  - 6.5.1.10. Serum AST less than 3 times the upper limit of normal
  - 6.5.1.11. Total Bilirubin less than 1.5 times the upper limit of normal
  - 6.5.1.12. Creatinine  $<1.5$  times the upper limit of normal
  - 6.5.1.13. Recovered from the acute toxic effects of all prior chemotherapy based on the enrolling physician's assessment (if some effects of chemotherapy are expected to last long term, patient is eligible if meeting other eligibility criteria).
  - 6.5.1.14. Weight greater than 12 kg
- 6.5.2. Procurement Exclusion Criteria
- 6.5.2.1. Rapidly progressive disease
  - 6.5.2.2. History or hypersensitivity to murine protein-containing products
  - 6.5.2.3. Tumor causing airway obstruction
  - 6.5.2.4. Currently receiving immunosuppressive drugs such as corticosteroids<sup>\$</sup>, tacrolimus or cyclosporine
  - 6.5.2.5. Severe previous toxicity from cyclophosphamide or fludarabine based on the enrolling physician's assessment
  - 6.5.2.6. HIV infection
- 6.5.3. Treatment Inclusion Criteria\*
- 6.5.3.1. Relapsed or refractory high-risk neuroblastoma
  - 6.5.3.2. Life expectancy of at least 12 weeks
  - 6.5.3.3. Age greater than 1 year and less than 21 years old
  - 6.5.3.4. Karnofsky/Lansky score of 60% or greater (**Appendix A**)
  - 6.5.3.5. Patients must have an ANC  $\geq 500/\mu\text{l}^{\#}$ , platelet count  $\geq 20,000/\mu\text{l}$ . Patients may be transfused to obtain a platelet count  $\geq 20,000/\mu\text{l}$ .
  - 6.5.3.6. Pulse Ox  $\geq 90\%$  on room air
  - 6.5.3.7. Serum AST less than 3 times the upper limit of normal
  - 6.5.3.8. Total Bilirubin less than 1.5 times the upper limit of normal
  - 6.5.3.9. Creatinine  $<1.5$  time the upper limit of normal
  - 6.5.3.10. Recovered from the acute toxic effects of all prior chemotherapy based on the enrolling physician's assessment (if some effects of chemotherapy are expected to last long term, patient is eligible if meeting other eligibility criteria).
  - 6.5.3.11. Absence of human anti-mouse antibodies (HAMA) prior to enrollment for patients who have received prior therapy with murine antibodies
  - 6.5.3.12. Patients must have autologous transduced NKTs with  $\geq 20\%$  expression of GD2-specific CAR.
  - 6.5.3.13. Informed consent and assent (as applicable) obtained from parent/guardian and child.
  - 6.5.3.14. Weight greater than 12 kg

#### 6.5.4. Treatment Exclusion Criteria

- 6.5.4.1. Rapidly progressive disease
- 6.5.4.2. Currently receiving any investigational drugs
- 6.5.4.3. History or hypersensitivity to murine protein-containing products
- 6.5.4.4. Cardiomegaly or bilateral pulmonary infiltrates on chest radiograph or CT. However, patients with cardiomegaly on imaging may be enrolled if they have an assessment of cardiac function (i.e., ECHO or MUGA) within 3 weeks of starting protocol therapy that is within normal limits. Additionally, patients with bilateral pulmonary infiltrates on imaging may be enrolled if the lesions are not consistent with active neuroblastoma (i.e., negative on functional imaging with PET or MIBG, or by pathologic assessment).
- 6.5.4.5. Tumor potentially causing airway obstruction
- 6.5.4.6. Pregnancy or lactation or not willing to use birth control
- 6.5.4.7. Currently receiving immunosuppressive drugs such as corticosteroids<sup>§</sup>, tacrolimus or cyclosporine
- 6.5.4.8. Severe previous toxicity from cyclophosphamide or fludarabine based on the enrolling physician's assessment
- 6.5.4.9. HIV infection

*\* All labs must be collected within 10 days prior to initiation of study-related treatment (except for verification of GD2 transduction)*

<sup>#</sup>: ANC  $\geq$  500 without the use G-CSF or GM-CSF for at least 48 hrs.

<sup>§</sup>: Patients may receive treatment if treated with corticosteroids with dose of less than 0.5 mg/kg/day of prednisone equivalent.

## 7. Immunotherapy Supply

### Generation of Transduced NKT Cells

Autologous Natural Killer T cells (NKT) expressing the 14G.28.IL-15 construct will be generated by retroviral transduction and ex vivo expansion. All of these procedures will be performed in our GMP facility as dictated by Standard Operating Protocols (SOP). Brief summaries of procedures are given below:

#### 7.1. Source Material

Patient will undergo leukocyte apheresis (leukapheresis). The leukapheresis product is used to generate the cellular components listed below for treatment and samples for infectious disease testing.

#### 7.2. Selection of Peripheral Blood NKTs by Immunomagnetic Bead Selection

Peripheral blood mononuclear cells (PBMC) will be isolated by density gradient separation after leukapheresis. PBMC are then subjected to magnetic-activated cell sorting (MACS) using Anti-iNKT MicroBeads (Miltenyi Biotec) using the Miltenyi CliniMACS Plus device. The positive fraction cells (NKT) will be mixed with irradiated artificial Antigen Presenting Cells (aAPCs; clone A4-4) at a ratio of two (2) NKT cells to greater than, or equal to one ( $\geq 1$ ) aAPCs, in Advanced RPMI-1640 culture medium supplemented with 9% (v/v) FBS, 1x GlutaMAX (NKT medium) and containing 100 ng/mL alpha-galactosyl-ceramide ( $\alpha$ GalCer), 200 IU/mL recombinant human interleukin (IL)-2 and 10 ng/mL recombinant human IL-21. The remaining negative fraction cells are cryopreserved for secondary NKT stimulation. Cells will be cultured in G-Rex™10 (Wilson Wolf) until transduced. Culture medium (NKT medium)

refreshment is performed as required and IL-2 and IL-21 added on an every-other-day basis.

### 7.3. **Retroviral Vector Production**

A retroviral producer line clone is generated for the 14G.28.IL-15 construct and a master cell-bank is generated according to our SOPs. The master-cell bank will be tested to exclude generation of replication competent retrovirus and infection by mycoplasma, HIV, HBV, HCV and others, according to our SOP.

Producer lines are grown to confluence. Supernatant is then harvested, filtered, aliquoted and rapidly frozen and stored at  $-80^{\circ}\text{C}$ . All batches of retroviral supernatant will be tested again in particular to exclude RCR and issued with a Certificate of Analysis (CoA) as directed by our SOPs.

### 7.4. **Retroviral Transduction and Secondary Stimulation**

- 7.4.1. As per the GINAKIT2 SOP, NKT cells will be transduced with the 14G.28.IL-15 retroviral vector and expanded in culture. During this phase of NKT expansion the assessment of cell numbers, NKT purity and transduction efficiency will guide the final disposition of the product. If numbers are insufficient and / or rate of expansion is low, CAR NKTs may be re-stimulated (secondary stimulation) with  $\alpha\text{GalCer}$ -loaded autologous PBMCs in an appropriate G-Rex<sup>TM</sup> device in NKT medium supplemented with IL-2 and IL-21. It is expected that by day 10-20 of culture, measures (CAR<sup>+</sup>NKT numbers, NKT purity and transduction efficiency) will be sufficient to proceed with cryopreservation of the product.
- 7.4.2. The decision to cryopreserve will be contingent upon achieving sufficient CAR<sup>+</sup>NKT numbers, NKT purity and transduction efficiency.
- 7.4.3. If sufficient CAR<sup>+</sup>NKT numbers are not achieved prior to day 20, then the product may be cultured beyond day 20, and cryopreserved at this later time point.
- 7.4.4. If the purity of the NKT product does not achieve release criterion ( $\geq 70\%$ ), the option to repeat the enrichment of the NKT using the Miltenyi Anti-iNKT MicroBeads and CliniMACS device, may be exercised. After enrichment, a period of ex vivo expansion culture may be performed, as per SOP. A reassessment will be made of CAR<sup>+</sup>NKT numbers, NKT purity and/or transduction efficiency and will guide the final disposition of the product.
- 7.4.5. In the event that CAR<sup>+</sup>NKT cell numbers are not sufficient after (7.4.5.), continued cell culture in medium supplemented with IL-2 and IL-21 may be performed, as per SOP, until CAR<sup>+</sup>NKT cell numbers are sufficient.
- 7.4.6. If cell proliferation is less than expected, the option to repeat stimulation of NKT with irradiated autologous PBMC loaded with  $\alpha\text{GalCer}$  in NKT medium supplemented with IL-2 and IL-21, as per SOP, may be exercised.
- 7.4.7. If the transduction efficiency of the NKT does not achieve release criterion ( $\geq 20\%$ ), the option to repeat the transduction of NKT, as per SOP, may be exercised.
- 7.4.9. When cell numbers, NKT purity and transduction efficiency are sufficient, the product will be cryopreserved, as per SOP.

### 7.5. **Freezing**

Cells are to be cryopreserved for storage prior to infusion. Cryopreservation is performed according to the appropriate GMP Facility SOP. QA/QC testing performed during the process, will evaluate the final product for sterility (bacterial and fungal), mycoplasma and for the presence of replication competent retroviruses (RCR) by PCR. Cells will be counted and the proportion of NKT cells (NKT purity) and CAR<sup>+</sup>NKT cells (transduction efficiency) determined by flow cytometry. In addition, the functionality of the product will be assessed in an in vitro luciferase-based, cytotoxicity assay against GD2<sup>+</sup> tumor cells. Upon QA/QC review a Certificate of Analysis for product release will be generated.

#### 7.6. Handling of Cells Prior to Administration

Cells will be cryopreserved according to institutional SOP. Prior to infusion, the autologous cells will be removed from liquid nitrogen storage according to GMP Facility SOP, transported to the patient bedside in liquid nitrogen according to GMP Facility SOP and thawed immediately prior to infusion according to GMP Facility SOP. Infusion is performed by the designated clinician and clinical staff.

### 8. Treatment Plan

#### Treatment Overview

| Months                          | At least 1 month prior to anticipated NKT cell infusion date |   |    |    |    |    |    |
|---------------------------------|--------------------------------------------------------------|---|----|----|----|----|----|
| Days                            |                                                              |   | -4 | -3 | -2 | -1 | 0  |
| Consent for NKT cell Generation | X                                                            |   |    |    |    |    |    |
| Obtain Blood Sample             | X                                                            |   |    |    |    |    |    |
| G28z.15 NKT Generation          | X                                                            |   |    |    |    |    |    |
| QA/QC, Release Testing          |                                                              | X |    |    |    |    |    |
| Consent for treatment           |                                                              |   | X  |    |    |    |    |
| Cyclophosphamide                |                                                              |   | X  | X  |    |    |    |
| Fludarabine                     |                                                              |   | X  | X  | X  |    |    |
| G28z.15 NKT Infusion            |                                                              |   |    |    |    |    | X* |

See Patient Evaluation Table in Section 9.4 for study-related procedures.

\*: CAR NKTs may be infused between Day 0 and Day +3.

**8.1.** Dose levels for autologous G28z.15 NKT cell immunotherapy. Dosing will be based on the actual number of transduced cells. All doses are per m<sup>2</sup>.

- Dose Level 1 =  $3 \times 10^6$
- Dose Level 2 =  $1 \times 10^7$
- Dose Level 3 =  $3 \times 10^7$
- Dose Level 4 =  $1 \times 10^8$
- Dose Level 5 =  $3 \times 10^8$
- Dose Level 6 =  $1 \times 10^9$

**8.2. Administration of Lymphodepleting Chemotherapy:**

Cyclophosphamide (500 mg/m<sup>2</sup>/day x 2 days) and fludarabine (30 mg/m<sup>2</sup>/day x 3 days) are the agents of choice, as there is the most experience with the use of these agents in facilitating adoptive immunotherapy. Cyclophosphamide and fludarabine will be given daily starting Day -4 prior to administration of G28z.15 NKTs. Potential side effects of cyclophosphamide and fludarabine are listed in **Appendix B**.

8.2.1. **Anti-emetics:** Anti-emetics in appropriate dosage for each patient will be prescribed as necessary before chemotherapy infusions. Corticosteroids should not be used as anti-emetics on this study.

8.2.2. **Supportive Care:** Use of supportive care agents such as blood products, PJP prophylaxis, and growth-factor support should be administered by the treating team as required for good patient care.

**8.3 Administration of G28z.15 NKTs:**

Patients will be evaluated in the clinic and eligibility and informed consent and assent (as applicable) obtained/verified. The cells will be cross-checked using patient specific identifiers and infused per our SOPs. Infusion administration and post-infusion monitoring will follow CAGT established SOPs. Patients will be monitored for clinical toxicity by standard NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 4.X at <http://ctep.cancer.gov> with the exception of Cytokine Release Syndrome (CRS) and neurological toxicities which will be graded according to the tables in **Appendix D**. A time period of 4 weeks will constitute the time for clinical safety monitoring.

Each subject will receive a dose of G28z.15 NKTs at the specified dose level. If no dose-limiting toxicities are detected, a maximum of three additional doses of G28z.15 NKTs may be given on the same dose level. Patients will need to meet treatment eligibility at the time of each start of lymphodepletion for reinfusions.

8.3.1. **Anti-emetics:** Anti-emetics in appropriate dosage for each patient will be prescribed as necessary for G28z.15 NKT cell infusions.

8.3.2. **Infusion of CAR NKTs:** G28z.15 NKTs will be given by intravenous infusion over up to 10 minutes and the IV flushed with saline on any day within the timeframe of

Day 0 to Day +3. The volume of infusion will depend upon the concentration of the cells when frozen, the dose, and the size of the patient. Monitoring will be undertaken according to institutional standards. All infusions will be given at Texas Children's Hospital.

- 8.3.3. Only products that meet study specific release criteria, as detailed on the CoA, will be infused. Frozen cells that do not meet all release criteria will not be released for infusion.
- 8.3.4. Additional Doses of G28z.15 NKT cells: Patients with a demonstrated response and/or at least stable disease at 4 weeks evaluation AND had no evidence of a DLT with the original infusion are eligible to receive additional doses of G28z.15 NKT cells at their original dose level (maximum of three additional doses) with lymphodepletion, if available. additional dose will consist of approximately the same number of cells as the first infusion depending on product availability. Patients will need to meet treatment eligibility at the time of each start of lymphodepletion for reinfusions.

#### **8.4. Supportive Care Guidelines**

- 8.4.1. Venous Access:  
An indwelling venous access catheter is not required for treatment on this protocol.
- 8.4.2. Allergic Reactions:  
Although no severe allergic reactions have been documented after the use of autologous GD2 T cells, the following agents should be available at the bedside prior to each injection.  
Hydrocortisone 2 mg/kg (max dose of 200 mg) IV, diphenhydramine 1 mg/kg IV (max dose of 50 mg), and epinephrine 0.01 mL/kg (of 1:1,000 solution, max dose of 0.5 mL) subcutaneously.
- 8.4.3. Fever and Neutropenia: Standard of care should be administered for complications resulting from neutropenia. Patients with fever will be admitted, blood (and other samples if indicated) will be sent for culture and broad-spectrum antibiotics will be administered according to the Texas Children's Hospital's evidence-based Fever and neutropenia guidelines.
- 8.4.4. Blood Products: Blood products should be given when appropriate.
- 8.4.5. Concomitant Therapy: No other cancer chemotherapy or immunomodulating agents will be used except as described below.
- 8.4.6. Cytokine Release Syndrome (CRS) and Neurotoxicity:  
Cytokine release syndrome and neurotoxicity are known side effects in up to 50% or more patients who received T cells expressing CARs containing additional costimulatory endodomains (71, 93, 101, 106, 107). This syndrome has not been described with non-genetically modified NKT cell therapies, but there is no prior

clinical experience with NKT cells expressing CARs. Patients will be closely monitored clinically to allow appropriate and timely intervention. We will collect serum before and after G28z.15 NKT cell infusion for cytokine monitoring, and continue to monitor subjects for 3 hours after infusion. Patients will then be monitored by frequent telephone contact or physical examination for 4 weeks by their treating physicians for evidence of incipient "cytokine storm" (onset of fever, malaise and dyspnea) or other serious adverse events and treated promptly. Management and monitoring (including CRP and cytokines) of cytokine storm is described in more detail in SOP F 05.11.XX and includes treatment options based on the clinical severity of the symptoms such as oxygen, inotropic agents, IL-6 receptor antibody (tocilizumab, 4-8 mg/kg/dose), IL-1 receptor antagonist (anakinra, 1-2mg/kg/dose), neutralizing TNF- $\alpha$  antibody (infliximab, 5-10 mg/kg/dose), and/or steroids (1-2 mg/kg/day of methylprednisolone or equivalent).

8.4.7. Pain Management: Monoclonal antibodies targeting GD2 can cause severe pain requiring analgesics. Patients who will develop pain will be assessed by objective pain score system (0-10, where 0 is no pain and 10 is the greatest pain the patient can possibly imagine/feel) or by FACES scale for children who cannot quantify their pain. Treating physician will decide the appropriate choice of analgesics and may include oral or i.v. medications and non-pharmacologic interventions.

8.4.8. Management of Hyperleukocytosis: In the absence of evidence of an infectious etiology, in case of leukocytosis patients will be admitted for close monitoring. If the WBC or ALC increases by greater than 5-fold / 24hrs after reaching normal values for age OR exceeds three times the upper limit of normal value for age, management may include aggressive hydration (up to 2-4 times normal maintenance) steroids (1-2 mg/kg/day of methylprednisolone or equivalent, leukapheresis and/or cytoreductive chemotherapy. (We will consider the combination of cyclophosphamide at 500mg/m<sup>2</sup>/dose and fludarabine at 30mg/m<sup>2</sup>/dose OR etoposide 75mg/m<sup>2</sup>/dose). Additional supportive measures will be implement per standard of care.

## 9. Study Evaluations

### 9.1. Assessment of Safety:

9.1.1. Toxicities will be graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 4.X at <http://ctep.cancer.gov>.

9.1.2. Assessment and grading of CRS and Neurotoxicity: CRS will be graded according to published guidelines by Lee et al (98) and grading tables are shown in **Appendix D**.

### 9.2. Immune Function Studies

Peripheral blood will be collected at the specified time points and used for all or some of the following studies depending on cell numbers and availability of reagents. Peripheral blood used for standard and research labs will not exceed 60

mL/day for adults and 3 mL/kg/day for children. We will isolate peripheral blood lymphocytes (PBMCs) as well as plasma from the samples:

9.2.1. PBMCs will be used for:

- a. Immune function studies including analysis of the frequency and phenotype and persistence of G28z.15 NKTs using flow cytometry (CAR, CD3/6B11/CD4/CD62L).
- b. If detection G28z.15 NKT cells is greater than 0.5% and there is persistent G28z.15 NKT cell expansion 3 months after T-cell infusion we will perform clonality studies (an extra 5 ml of blood). Further evaluations may be done based on clinical decisions.
- c. Evaluate the frequency of NKT target cells: CD1d+CD33+CD14+CD163+ M2-like myelomonocytic cells in blood and in tumor biopsies and BM aspirates (when available). As BM is the primary site of NB metastases, we will examine whether infiltration with CAR NKTs correlates with decreased frequency of GD2+ NB cells in BM. In addition, we will analyze frequency and activation status (CD69) of NK cells (CD3-CD56+), which are known to be activated by NKTs and are in part responsible for their anti-metastatic effect (108).

9.2.2. Plasma will be used to determine:

- a. Antibody generation to the CAR-NKT cell product.
- b. The concentration of molecules that modulate immune responses such as cytokines or chemokines using Elisa and multiplex assays.

9.2.3. RCR testing by PCR. Samples for RCR testing will be collected per SOP J 02.51.XX.

9.2.4. If insufficient PBMCs are obtained to perform all of the above studies, assays will be prioritized as follows: RCR testing > CAR NKT frequency/persistence and other testing by flow cytometry> Cytokine level measurement > Cytotoxicity assays.

9.2.5. If a patient's hemoglobin is less than 7.0 g/dL at any of the evaluation times, the amount of blood drawn for the evaluation will be reduced and may be obtained over more than one venipuncture, if necessary. Studies will be conducted depending on the availability of the patient and the ability to safely draw the amount of blood needed for the studies. The time points given are approximate as patients may not always be able to keep appointments. Every effort will be made, however, to obtain studies on the above schedule. Residual samples will be stored for any future study-related assays.

9.3. **Disease Evaluation:**

9.3.1. Although response is not the primary endpoint of this trial, patients with measurable disease will be assessed by the International Neuroblastoma Response Criteria (109). Evaluations of tumor burden will be performed within 4 weeks prior to start of lymphodepletion (preferably within 2 weeks) and 4 weeks (range 4-6 weeks) after the infusion. Note: Catecholamines will not be obtained as part of this study and therefore not used to assess response as noted in 9.3.2.

**9.3.2. International Recommendations for Neuroblastoma Response Criteria  
(see Appendix C for the complete criteria)**

**9.3.2.1. Response Criteria for Morphologic Bone Marrow Only Disease:**

Only those patients with morphologic evidence of neuroblastoma by routine H and E staining will be evaluable to assess bone marrow response.

- a. Complete Response (CR): No tumor cells detectable by routine morphology on two subsequent bilateral bone marrow aspirates and biopsies done at least three weeks apart after study entry.
- b. Progressive Disease (PD): Tumor seen on morphology on two consecutive bone marrows done at least three weeks apart in patients who had NO tumor in bone marrow at study entry. Patients who enter study with bone marrow tumor by morphology will be considered to have progressive disease if there is a minimum of 25% tumor in the marrow by morphology AND a doubling in the amount of tumor in the marrow compared to the level present at study entry. (For example, a patient entering with 5% tumor in marrow by morphology must increase to >25% tumor to have PD; a patient entering with 30% tumor must increase to >60% tumor).
- c. Stable Disease (SD): Will be defined as persistence of an amount of tumor in the bone marrow by morphology that does not meet criteria for either complete response or progressive disease. Patients who clear morphologic tumor but have immunocytologic tumor will be called SD.

**9.3.2.2. Response Criteria for Evaluable Disease Measured by MIBG/PET/  
or Bone Scan Positive Lesions**

Patients who have a positive nuclear medicine scan at the start of therapy will be evaluable for response. The following will be used to evaluate response:

- a. Complete Response: Complete resolution of all positive lesions
- b. Partial Response: Resolution of at least one positive lesion, with persistence of other positive lesions.
- c. Stable Disease: No change in the number of positive lesions (includes patients who have same number of positive lesions but changes in intensity).
- d. Progressive Disease: Development of new MIBG positive lesions AND / OR bone marrow involvement detected in a patient with previous negative bone marrow finding.

**9.3.3. Imaging:** Disease evaluation studies include diagnostic imaging to document measurable disease and response to therapy (CT scans, MRI, MIBG scan, PET scan or bone scan) based on section 9.3. However, the selection of studies will be individualized based on which studies have been most informative in the particular patient. If diagnostic imaging studies are performed at other times either during or after treatment on this study, that data will be requested and information gained will be used for this study.

- 9.3.4. **Bone Marrow Evaluation:** To evaluate bone marrow involvement by neuroblastoma, the presence or absence and the effect of infused cells before and after treatment, each patient will have bilateral bone marrow aspirates and biopsies performed within 4 weeks prior to start of lymphodepletion, at 2 weeks, and at 4 weeks (range 4-6 weeks for response to therapy sample) after treatment. Bone marrow evaluation after CAR NKT cell infusion will be only performed if patient had disease involvement in the bone marrow at study entry. Additional evaluations may be done as clinically indicated.

#### 9.4. Patient Evaluation - Monitoring Related to G28z.15 NKT Cell Infusions

| Study                                                  |     |       |                |     |     |     |     |     |     |     |     |      |                          |                             |
|--------------------------------------------------------|-----|-------|----------------|-----|-----|-----|-----|-----|-----|-----|-----|------|--------------------------|-----------------------------|
|                                                        | Pre | D - 4 | D 0            | W 1 | W 2 | W 3 | W 4 | W 8 | M 3 | M 6 | M 9 | M 12 | Every 6 months x 4 years | Annually X 15 years (total) |
| Hx†                                                    | X   | X     | X              | X   | X   | X   | X   | X   | X   | X   | X   | X    | X                        | X                           |
| PE†                                                    | X   | X     | X              | X   | X   | X   | X   | X   | X   | X   | X   | X    | X                        | X                           |
| Weight/Height                                          | X   | X     |                |     |     |     | X   |     |     |     |     |      |                          |                             |
| Vital Signs                                            | X   | X     |                | X   | X   | X   | X   | X   |     |     |     |      |                          |                             |
| Pulse Oximetry                                         |     | X     | X              |     |     |     |     |     |     |     |     |      |                          |                             |
| Performance Status                                     | X   | X     |                |     |     |     |     |     |     |     |     |      |                          |                             |
| Urine Pregnancy Test if post menarche                  | X   |       |                |     |     |     |     |     |     |     |     |      |                          |                             |
| CBC d/p                                                | X   | X     | X              | X‡  | X‡  | X‡  | X   | X   | X   | X   | X   | X    | X                        | X                           |
| Chem 10                                                | X   | X     | X              | X   | X   | X   | X   | X   |     |     |     |      |                          |                             |
| AST/Total Bili                                         | X   | X     | X              | X   | X   | X   | X   | X   |     |     |     |      |                          |                             |
| CT/MRI C/A/P*                                          | X   |       |                |     |     |     | X   |     |     |     |     |      |                          |                             |
| CT/MRI of other know sites of disease* &               | X   |       |                |     |     |     | X   |     |     |     |     |      |                          |                             |
| Tc <sup>99</sup> Bone Scan and/or MIBG and/or PET*, \$ | X   |       |                |     |     |     | X   |     |     |     |     |      |                          |                             |
| Bilat BMA/Bx if bone marrow positive at study entry**  | X   |       |                |     | X%  |     | X   |     |     |     |     |      |                          |                             |
| CAR NKT Immune Monitoring and Safety Studies           | X   |       | X <sup>#</sup> | X   | X   | X   | X   | X   | X   | X   | X   | X    | X                        | X                           |
| HLA Typing                                             | X   |       |                |     |     |     |     |     |     |     |     |      |                          |                             |

† Starting Month 6, status updates obtained by documented contact with the patient/parent may substitute history and physical exams if the patient cannot have a scheduled visit with their provider at the pre-determined time points.

\* Initial imaging for disease evaluation must be obtained within 4 weeks (preferably 2 weeks) prior to start of lymphodepletion and 4 weeks (range 4-6 weeks) post-infusion. Additional evaluations can be done as clinically indicated.

\*\* Bone marrow assessment for disease evaluation must be obtained within 4 weeks (preferably 2 weeks) prior to start of lymphodepletion, and at 2 weeks, and at 4 weeks (range 4-6 weeks) post-infusion. Additional evaluations can be done as clinically indicated.

\$: CT/MRI of other known site in addition to the primary site is at the discretion of the PI.

§: Selection of studies will be individualized based on which studies have been most informative in the particular patient.

‡ CBC with diff will be measured 3 times per week for Week 1 to 3.

# Blood will be drawn pre-infusion and again 3 hours after the infusion (the post-infusion sample is just for quantitative real-time PCR).

%: At Week 2 post-infusion, bilateral bone marrow aspirate/biopsies and/or tumor biopsies may be obtained. Tumor biopsies should be preferred in patients without bone marrow involvement. In patient with bone marrow involvement, both biopsies may be considered.

Note: Day 0 is always the day of CAR NKT cell infusion, CAR NKTs may be infused between Day 0 and Day +3.

If patients receive multiple infusion, time for monitoring patients will be re-started from every infusion.

#### 9.4.1. Pre-Study Evaluations

##### 9.4.1.1. Pathology-proven refractory or relapsed neuroblastoma

##### 9.4.1.2. Laboratory Tests

- a. CBC with differential and platelets
- b. Chem 10
- c. AST, Total Bilirubin
- d. Bone Marrow Aspirate (+/- biopsy)
- e. Immune evaluation
- f. Pregnancy test for females of child bearing potential
- g. HLA typing will be performed to ensure autologous identity of T cell product

##### 9.4.1.3. Radiographic Disease Evaluation

- a. CT/MRI of primary and metastatic locations
- b. MIBG scan, PET scan or Bone Scan (based upon patient history)

##### 9.4.1.4. Biopsy Samples (Slides and Tissue Scrolls) if available

- a. 5-10 unstained slides from primary or most recent biopsy sample
- b. At least one 40 micron tissue scroll from the primary or most recent biopsy sample

9.4.2. History and Physical Examination: Full history and physical examination will be done pre-infusion and at 1, 2, 3, 4, and 8 weeks post-infusion and then every three months for a year, every 6 months for 4 years, and then yearly for a total of 15 years. A summary of the patient's history may be obtained by a research coordinator/nurse/MD starting at the 6 month time frame if the patient does not have or cannot schedule a visit with their primary oncologist/MD to coincide with this timeframe.

9.4.3. Routine Laboratory Investigations: The following routine laboratory investigations will be obtained as outlined in the study calendar:

- a. CBC with differential and platelets: Day -4, Day 0, three times per week on Week 1 to 3, and then once a week on Week 4 during the first 28-day toxicity period (Section 9.4. of the clinical protocol) or at least once per day if patient is admitted. Followed by additional monitoring on Week 8, Months 3, 6, 9, 12; every 6 months for additional 4 years and annually for additional 10 years (monitoring ends at 15 years follow up from infusion).
- b. Chem 10: pre-infusion, at 1, 2, 3, 4, and 8 weeks post-infusion
- c. Total Bilirubin, Serum AST: pre-infusion, at 1, 2, 3, 4, and 8 weeks post-infusion
- d. Pregnancy test: pre-infusion for females who have achieved menarche.

9.4.4. Tests of Function, Persistence and Safety: Blood will be collected at the time points indicated in the patient evaluation calendar (Up to 60 mL/day for adults and 3 mL/kg/day for children). The following analysis will be performed on peripheral

blood to monitor function, immunologic effect, persistence, and safety of transduced NKT-cells at time points indicated in the study calendar:

- 9.4.4.1. Flow cytometry analysis for the frequency of G28z.15 NKT cells, CD1d+CD33+CD14+CD16+/- myelomonocytic cells, NK cells (CD3-CD56+) and Tregs.
- 9.4.4.2. Quantitative real-time PCR for G28z.15.
- 9.4.4.3. RCR testing by PCR. RCR testing will be performed pre-study, 3, 6, 12 months. If all post treatment samples are negative during the first year, the subsequent samples will be archived yearly for a total of 15 years. Aliquot of cells and serums will also be archived for use in future studies for RCR as required by the FDA or RAC.
- 9.4.4.4. Serum collection for HAMA testing and serum cytokine levels.
  - a. On day of infusion (between Day 0 and Day +3) two separate samples will be taken:
    - i. Before infusion, blood will be drawn for testing as outlined in Section 9.4
    - ii. Three hours after infusion, a small additional sample (5-10 mL) will be drawn. This sample is solely for quantitative real-time PCR for G28z.15.
    - iii. Serum from blood drawn for functional studies on day 0, week 4, and week 12 will be stored for measurement of human anti-mouse or rat antibodies (HAMA). These studies will be performed in the event of a suspected immunologic reaction.
- 9.4.5. Diagnostic Imaging: Routine studies for documentation of tumor size 2-4 weeks before start of lymphodepletion and 4 weeks (range 4-6 weeks) after CAR NKT infusion are required. If diagnostic imaging studies are performed at other times either during or after treatment on this study, those data will be collected and information gained will be used for this study.
  - a. CT or MRI of Chest/Abdomen/Pelvis
  - b. CT/MRI of any other known sites of disease (if clinically indicated)
  - c. Bone scan, MIBG scan, and/or PET scan (if clinically useful per patient's disease history).
- 9.4.6. Other tissues
  - 9.4.6.1. Bilateral bone marrow aspirations and +/- biopsies within 4 weeks prior to start of lymphodepletion, at 2 weeks, and at 4 weeks (range 4-6 weeks) after CAR NKT infusion (if positive prior to infusion or evidence of progression of tumor by other evaluations).
  - 9.4.6.2. Tumor biopsy shall be obtained at 2 weeks after CAR NKT cell infusion.
  - 9.4.6.3. If tumor tissue or bone marrow is required at any time after treatment, a sample will be requested to assess the presence of transduced NKT-cells within the tumor or metastatic sites (e.g. bone marrow).
  - 9.4.6.4. For any patients who develop a second neoplasm, significant hematologic or neurologic disorder during the trial, a biopsy sample of the neoplastic tissue will be assayed for RCR (if a sample can be obtained).
  - 9.4.6.5. If the patient dies, an autopsy will be requested. If granted, tissue will be requested to assess presence of transduced cells.

## 10. Off-Study Criteria

- 10.1 Death
- 10.2 Lost to follow-up
- 10.3 Completion of planned therapy and follow-up
- 10.4 Withdrawal of consent for any further data submission.

## 11. Statistical Analysis and Stopping Rules

The Phase 1 trial will determine the maximum tolerated dose (MTD), defined as the highest dose level that will have at most a 33% chance of inducing the following NKT-cell-related dose-limiting toxicities (DLTs) within 28 days after the first infusion of NKTs. If the patient receives more than one CAR NKT infusion, safety assessments on additional doses will be continued until 28 days post-additional infusion, but toxicities after the additional doses will not be considered DLTs.

Toxicities will be graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 4.X at <http://ctep.cancer.gov> with the exception of Cytokine Release Syndrome (CRS) and neurological toxicities which will be graded according to the tables in **Appendix D**.

Dose-limiting toxicities (DLTs):

- a. A DLT is defined as any hematological or non-hematological Grade 3-5 toxicity that is considered to be probably related to the CAR NKT cells.
- b. CRS and neurological toxicities grade 3 and 4, expected reactions seen with the use of CAR-based immunotherapy such as fever or hypotension lasting less than 72 hours, will not be considered DLTs. Those symptoms lasting >72 hours will be considered DLT.
- c. Any other unexpected grade 3 or greater toxicity felt to be related to or resulting from the CRS or due to neurological toxicity is included in the definition of DLT.
- d. Grade 3 and 4 CRS or neurological toxicities that are persistent beyond 72 hours will be reported to the FDA in an expedited fashion and will be considered DLTs.

This study will implement the traditional 3+3 design, which is conservative and requires no modeling of the dose-toxicity curve. The trial operates with cohorts of three patients. The first three patients will be treated at the lowest dose level. There will be at least 1-week gap between each enrollment on a dose level until the MTD is identified. If none of the three patients has a DLT, another three patients will be treated at the next higher dose level. However, if one of the three patients experiences a DLT, then three more patients will be treated at the same dose level. If at least two patients among a cohort of three to six patients experience DLTs, then the trial will be halted, and the dose level just below this toxic dose level will be defined as the MTD level. If none of the dose levels has more than one patient who has DLT, then the highest dose level will be defined as the MTD. An additional safety provision is that a total of 6 patients will be studied at the MTD level. The maximum sample size for six dose levels is 36. If there is no DLT, then the expected sample size is 21.

### 11.1. Early Termination of Study Participation

- 11.1.1. Depending on patient availability, a maximum of 36 patients can be accrued into this Phase I trial.
- 11.1.2. If the patient/parent desires to withdraw from the study or if the physician feels that it is in the best interest of the patient, treatment will be discontinued.

### 11.2. Stopping Rules

- 11.2.1. If a patient dies within 30 days of G28z.15 NKT cell infusion of any cause except disease progression, accrual will be paused to allow consultation with the FDA.
- 11.2.2. If a patient has a Grade 4 CRS or neurotoxicity (including peripheral nerve injury from GD2-targeting related side effects) which does not resolve within 1 week of adequate therapy, the study will be paused to allow consultation with the FDA.

- 11.3. **Analysis of CAR<sup>+</sup> NKTs Expansion and Persistence.** The frequency of CAR<sup>+</sup> NKTs will be summarized at pre- and post-infusion time points using descriptive statistics (mean  $\pm$  SD, medians and ranges) to evaluate their expansion and persistence. Changes in each of these NKTs from pre-infusion to each time point of post-infusion will be assessed and compared using paired t-tests, or when appropriate, the Wilcoxon signed rank test. Scatter plots will be generated to graphically illustrate patterns of NKT cell expansion. Longitudinal modeling techniques like the random coefficient mixed model will be employed to model the repeated measurements to study their proliferation over time and estimate the magnitude of expansion or decline of NKTs. The validity of the normality assumption on these data will be examined and appropriate transformations will be considered. Dose level will be included as an independent variable in the model to account for different doses received by patients. Alternative strategies to analyze the immunologic measurement include the calculation of area under the curve over time for NKT cell frequencies or piecewise longitudinal models based on apparent trends from plots of growth curves.
- 11.4. **Analysis of Anti-tumor Activity.** Although it is not the primary endpoint, we will summarize tumor response by calculating overall response rates along with exact 95% binomial confidence intervals, and report the Kaplan-Meier curves for the relapse-free survival. We will summarize overall and disease-free survival in the entire patient cohort using Kaplan-Meier product-limit methods. Time to progression (TTP) will be analyzed by the cumulative incidence function using Gray's method (110). TTP will be defined as the time from first injection to disease progression per the INRC for patients with active disease at the time of G28z.15 NKT cell infusion, where death was the competing risk.
- 11.5. **Correlation of CAR<sup>+</sup> NKT cell Persistence and Anti-tumor Activity with Phenotyping of the Infused Cell Line.** The association of time to the last detectable CAR<sup>+</sup> NKT cell persistence with the phenotype of infused cell line will be analyzed by the Cox model. We will use the logistic model to analyze the association of tumor response with the phenotype of infused cell line.

## **12. Adverse Event Reporting Requirements**

### **12.1. Purpose**

Adverse event data collection and reporting, which are required as part of every clinical trial, are done to ensure the safety of patients enrolled in the studies as well as those who will enroll in future studies using similar agents. Certain adverse events must be reported in an expedited manner to allow for timelier monitoring of patient safety and care.

### **12.2. Records to be kept**

Database documenting the dates and doses of therapy as well as clinical chemistries, hematologic parameters, the clinical status and occurrence of any adverse events and subsequent interventions will be kept on all patients.

Records will be kept according to CAGT SOP J01.10.XX, Research Chart Components.

### **12.3. Reporting Requirements**

12.3.1. Register all patients with Cell and Gene Therapy Research Coordinator.

12.3.2. Enter all patients by phoning Dr. Heczey. The following forms should be completed:

- a. Eligibility checklist
- b. On study form

12.3.3. Drug Toxicity and/or Adverse Reactions

12.3.3.1. Adverse events will be collected as per CAGT SOP J 02.05.XX and J02.75.XX. Data on all adverse experiences/toxicities must be collected for 4 weeks after the dose of the study drug/biologic.

12.3.3.2. Reporting of CRS: Grade 3 or greater CRS will be reported expediently (Grade 4 if not resolved to at least Grade 3 in 72hrs; Grade 3 if not resolved to Grade 2 in two weeks) to the FDA. Grading will be based on published guidelines by Lee et al (98).

12.3.3.3. Serious adverse events will be collected and reported as per CAGT SOP J 02.06.XX. Reporting of serious adverse event related to gene transfer will continue for 15 years per FDA guidelines.

## **13. Informed Consent**

All patients and/or their legal guardian must sign a document of informed consent consistent with local institutional and Federal guidelines stating that they are aware of the investigational nature of this protocol and of the possible side effects of treatment. Further, patients must be informed that no efficacy of this therapy is guaranteed, and that unforeseen toxicities may occur. Patients have the right to withdraw from this protocol at any time. No patient will be accepted for treatment without such a document signed by the legal guardian. Full confidentiality of patients and patient records will be provided according to institutional guidelines.

13.1. Before acceptance of patients for this protocol, the investigator(s) will explain specifically the following to the patient and/or parents:

13.1.1. The nature of a Phase I study.

13.1.2. There exists a chance that therapeutic benefits may be detected following treatment with autologous G28z.15 NKT cell infusions, but that no direct benefit may be seen.

13.1.3. That certain toxic side effects have been encountered in human subjects with the use of 2<sup>nd</sup>-generation chimeric antigen receptors similar to G28z.15 CAR expressed in NKTs studied in this protocol.

13.1.4. That these side effects have included systemic reactions, and that the first patient treated with autologous G28z.15 NKT cells at dose level 5 ( $3 \times 10^8$  cells/m<sup>2</sup>) died within 3 weeks on this study after found to have polyclonal hyperleukocytosis complicated by multiorgan dysfunction.

13.1.5. That blood samples will be obtained over a period of time before and after infusion of modified NKTs.

13.1.6. The potential risks and benefits of cyclophosphamide and fludarabine.

13.1.7. That a copy of the signed consent form will be furnished to the patient and/or parents.

If there is conflict between the parent's and child's wishes, patients will have an opportunity to discuss the proposal with a representative from the social services or psychology departments, who will serve as a patient advocate and who should also witness the consent.

## **14. Clinical Trial Oversight and Monitoring**

This protocol will be conducted in accordance with the Cell/Gene Therapy Monitoring Plan on file with the FDA.

This protocol will be monitored in accordance with the current NCI-approved Data Safety Monitoring Board Plan for investigator-initiated Phase I and II studies in the Dan L Duncan Comprehensive Cancer Center at Baylor College of Medicine.

The conduct of this clinical trial will be evaluated in accordance with the Texas Children's Cancer Center and Center for Cell and Gene Therapy Quality Assurance Policy and Procedure Plan and Quality Control Policy and Procedure Plan.

## 15. APPENDIX A: PERFORMANCE STATUS SCALES/SCORES

| <i>Performance Status Criteria</i>                                                |                                                                               |        |                                                                                                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------|
| <i>Karnofsky and Lansky performance scores are intended to be multiples of 10</i> |                                                                               |        |                                                                                                                     |
| Karnofsky                                                                         |                                                                               | Lansky |                                                                                                                     |
| Score                                                                             | Description                                                                   | Score  | Description                                                                                                         |
| 100                                                                               | Normal, no complaints, no evidence of disease                                 | 100    | Fully active, normal                                                                                                |
| 90                                                                                | Able to carry on normal activity, minor signs or symptoms of disease          | 90     | Minor restrictions in physically strenuous activity                                                                 |
| 80                                                                                | Normal activity with effort; some signs or symptoms of disease                | 80     | Active, but tires more quickly                                                                                      |
| 70                                                                                | Cares for self, unable to carry on normal activity or do active work          | 70     | Both greater restriction of and less time spent in play activity                                                    |
| 60                                                                                | Required occasional assistance, but is able to care for most of his/her needs | 60     | Up and around, but minimal active play; keeps busy with quieter activities                                          |
| 50                                                                                | Requires considerable assistance and frequent medical care                    | 50     | Gets dressed, but lies around much of the day; no active play, able to participate in all quiet play and activities |
| 40                                                                                | Disabled, requires special care and assistance                                | 40     | Mostly in bed; participates in quiet activities                                                                     |
| 30                                                                                | Severely disabled, hospitalization indicated. Death not imminent              | 30     | In bed; needs assistance even for quiet play                                                                        |
| 20                                                                                | Very sick, hospitalization indicated. Death not imminent.                     | 20     | Often sleeping; play entirely limited to very passive activities                                                    |
| 10                                                                                | Moribund, fatal processes progressing rapidly.                                | 10     | No play; does not get out of bed                                                                                    |

## 16. APPENDIX B: POTENTIAL SIDE EFFECTS OF LYMPHODEPLETING CHEMOTHERAPY

Risks and side effects related to CYCLOPHOSPHAMIDE include those which are:

| Likely                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Less likely                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rare but serious                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Loss of appetite</li> <li>• Nausea</li> <li>• Vomiting</li> <li>• Fewer white blood cells in the blood.</li> <li>• A low number of white blood cells may make it easier to get infections.</li> <li>• Hair loss</li> <li>• Decreased ability of the body to fight infection</li> <li>• Absence or decrease in the number of sperm which may be temporary or permanent which may decrease the ability to have children</li> </ul> | <ul style="list-style-type: none"> <li>• Abnormal hormone function which may lower the level of salt in the blood</li> <li>• Abdominal pain</li> <li>• Diarrhea</li> <li>• Fewer red blood cells and platelets in the blood</li> <li>• A low number of red blood cells may make you feel tired and weak.</li> <li>• A low number of platelets may cause you to bruise and bleed more easily.</li> <li>• Bleeding and inflammation of the urinary bladder</li> <li>• Absence or decrease monthly periods which may be temporary or permanent and which may decrease the ability to have children</li> <li>• Temporary blurred vision</li> <li>• Nasal stuffiness with IV infusions</li> <li>• Skin rash</li> <li>• Darkening of areas of the skin and finger nails</li> <li>• Slow healing of wounds</li> <li>• Infections</li> </ul> | <ul style="list-style-type: none"> <li>• Heart muscle damage which may occur with very high doses and which may be fatal.</li> <li>• Abnormal heart rhythms</li> <li>• Damage and scarring of lung tissue which may make you short of breath</li> <li>• A new cancer or leukemia resulting from this treatment.</li> <li>• Damage or scarring of urinary bladder tissue</li> <li>• Severe allergic reaction which can be life threatening with shortness of breath, low blood pressure, rapid heart rate chills and fever</li> <li>• Infertility which is the inability to have children</li> </ul> |

**Risks and side effects related to FLUDARABINE (IV) include those which are:**

| Likely                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Less likely                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rare but serious                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Loss of appetite</li> <li>• Nausea or the urge to vomit</li> <li>• Decreased number of red blood cells, white blood cells (neutrophil/granulocyte), and/or platelets (a blood cell that helps clot blood)</li> <li>• Muscle weakness of the whole body</li> <li>• Cough</li> <li>• Shortness of breath</li> <li>• Fatigue or tiredness</li> <li>• Fever</li> <li>• Infection</li> <li>• Pain</li> <li>• Increased risk of unusual infections lasting more than 6 months</li> </ul> | <ul style="list-style-type: none"> <li>• Skin rash with the presence of macules (flat discolored area) and papules (raised bumps)</li> <li>• Diarrhea</li> <li>• Irritation or sores in the lining of the mouth, voice box, throat, and windpipe</li> <li>• Vomiting</li> <li>• Peripheral Neuropathy - Commonly known as "pins and needles," where part of the body (typically a foot or hand) begins to tingle and becomes numb, or "falls asleep"</li> <li>• Blurred vision, double vision and/or loss of vision (blindness)</li> <li>• Fear of light</li> <li>• Inflammation of the lungs that may cause difficulty breathing and can be life-threatening</li> <li>• Chills</li> <li>• An increase in the number of a type of white blood cell (called eosinophils) in the blood</li> <li>• Agitation or restlessness</li> <li>• Confusion</li> <li>• Weakness or paralysis (loss of muscle function) caused by damage to peripheral nerves (those nerves outside of brain and spinal cord)</li> <li>• Inflammation (swelling and redness) or degeneration of the peripheral nerves (those nerves outside of brain and spinal cord) causing numbness, tingling, burning</li> <li>• Pain of the urinary tract</li> <li>• Inflammation (swelling and redness) of the paranasal sinuses which may or may not be a result of infection</li> </ul> | <ul style="list-style-type: none"> <li>• Severe rash with redness, pain and/or blisters. When pressure is applied to an area, the skin will detach from the lower layers.</li> <li>• A rare autoimmune disorder called Evan's syndrome in which the body makes antibodies that destroy the red blood cells, platelets and white blood cells</li> <li>• Sudden damage to the red blood cells (hemolytic anemia) which could cause a rapid decrease in the number of red blood cells such that you may be tired, weak, feel short of breath, and may require a blood transfusion</li> <li>• Coma and/or abnormal brain function</li> <li>• Convulsion or seizure</li> <li>• Blindness</li> <li>• A rare disorder that damages the material that covers and protects nerves in the <u>white matter of the brain</u>. The disorder may cause headaches, loss of coordination, clumsiness, loss of language ability, memory loss, vision problems, and weakness of the legs and arms that gets worse.</li> <li>• Inflammation (swelling and redness) of the bladder not due to urinary tract infection</li> <li>• Severe potentially life-threatening damage to the lungs which can lead to fluid in the lungs</li> <li>• Bleeding from the lungs</li> <li>• Kidney damage which may require dialysis</li> </ul> |

## 17. APPENDIX C: International Neuroblastoma Response Criteria

(International Recommendations)<sup>6</sup>

| SITE         | TEST                                                                                                                          | COMPLETE RESPONSE | VERY GOOD PARTIAL RESPONSE                        | PARTIAL RESPONSE                                  | MIXED RESPONSE                                                                                                                                           | STABLE DISEASE                                                              | PROGRESSIVE DISEASE                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Primary      | 3 dimensional CT or MRI imaging (determine volume from product of three dimensions physical exam and/or surgical measurement) | no tumor          | >90% reduction in 3-dimensional tumor volume      | 50-90% reduction in 3-dimensional tumor volume    |                                                                                                                                                          |                                                                             |                                                                                                   |
| Metastases   | Bone marrow <sup>1</sup> (aspirate x 2 and biopsy x 2)                                                                        | no tumor          | no tumor                                          | no or only one sample with tumor <sup>2</sup>     | 50-90% reduction of any measurable lesion (primary or metastases); no new lesions; <25% increase in any existing lesions, exclude bone marrow evaluation | no new lesions; <25% increase in any lesion; exclude bone marrow evaluation | any new lesion; increase of any measurable lesion by >25%; previous negative bone marrow positive |
|              | Bone x-rays and scintigraphy (Tc and/or MIBG)                                                                                 | no lesions        | all lesions improved; no new lesions <sup>3</sup> | all lesions improved; no new lesions <sup>4</sup> |                                                                                                                                                          |                                                                             |                                                                                                   |
|              | Liver imaging (ultrasound, CT, or MRI)                                                                                        | no tumor          | no tumor                                          | 50-90% reduction                                  |                                                                                                                                                          |                                                                             |                                                                                                   |
|              | Chest x-ray, chest CT scans if x-ray abnormal                                                                                 | no tumor          | no tumor                                          | 50-90% reduction                                  |                                                                                                                                                          |                                                                             |                                                                                                   |
|              | Physical exam <sup>5</sup>                                                                                                    | no tumor          | no tumor                                          | 50-90% reduction                                  |                                                                                                                                                          |                                                                             |                                                                                                   |
| Tumor marker | Urine catecholamine metabolites (HVA & VMA)                                                                                   | Normal            | normal or both decreased >90%                     | both decreased 50-90%                             |                                                                                                                                                          |                                                                             |                                                                                                   |

Response must be evaluated before and after surgery for the primary site. If complete response, very good partial response, or partial response is achieved surgically, indicate such when reporting the response.

The total response can be no better than the worst response in any subcategory (e.g., if primary = complete response, metastases = partial response, and, tumor marker = very good partial response, the total response = partial response).

1) Immunocytology results are not used to determine response.

2) One sample may be positive only if there is a reduction in the number of sites originally positive for tumor at diagnosis.

3) <sup>99</sup>Tc bone scan may show residual abnormalities but the MIBG scan (if performed) must be negative.

4) <sup>99</sup>Tc bone scan and/or MIBG scan (if performed) must show improvement, but residual abnormalities may be present on either scan.

5) Measure palpable lymph nodes in 3 dimensions and calculate tumor volume.

6) Brodeur G et al: JCO 1993; 11:1466-1477

## 18. APPENDIX D: Grading of CRS and Neurological Toxicities

### CRS Grading Scale

| Grade    | Symptoms                                                                                                                                                                                                                                                                                  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <ul style="list-style-type: none"> <li>• Symptoms are not life threatening and require symptomatic treatment only (e.g. fever, nausea, fatigue, headache, myalgia, malaise)</li> </ul>                                                                                                    |
| <b>2</b> | <ul style="list-style-type: none"> <li>• Symptoms require and respond to moderate intervention</li> <li>• Oxygen requirement &lt;40% or hypotension responsive to fluids or</li> <li>• low dose of one vasopressor or Grade 2 organ toxicity</li> </ul>                                   |
| <b>3</b> | <ul style="list-style-type: none"> <li>• Symptoms require and respond to aggressive intervention</li> <li>• Oxygen requirement <math>\geq</math> 40% or hypotension requiring high dose or multiple vasopressors or</li> <li>• Grade 3 organ toxicity or Grade 4 transaminitis</li> </ul> |
| <b>4</b> | <ul style="list-style-type: none"> <li>• Life-threatening symptoms</li> <li>• Requirements for ventilator support or Grade 4 organ toxicity (excluding transaminitis)</li> </ul>                                                                                                          |
| <b>5</b> | <ul style="list-style-type: none"> <li>• Death</li> </ul>                                                                                                                                                                                                                                 |

### Neurological Toxicity Grading Scale

| Grade    | Symptoms                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <ul style="list-style-type: none"> <li>• Somnolence-mild drowsiness or sleepiness</li> <li>• Confusion-mild disorientation</li> <li>• Encephalopathy-mild symptoms</li> <li>• Dysphasia-not impairing ability to communicate</li> </ul>                                                                                                        |
| <b>2</b> | <ul style="list-style-type: none"> <li>• Somnolence-moderate, limiting instrumental ADL*</li> <li>• Confusion-moderate disorientation, limiting instrumental ADL*</li> <li>• Encephalopathy-limiting instrumental ADL*</li> <li>• Dysphasia-moderate impairing ability to communicate spontaneously</li> </ul>                                 |
| <b>3</b> | <ul style="list-style-type: none"> <li>• Somnolence – obtundation or stupor</li> <li>• Confusion-severe disorientation, limiting self-care-ADL*</li> <li>• Encephalopathy-limiting self-care ADL*</li> <li>• Dysphasia-severe receptive or expressive characteristics, impairing ability to read, write or communicate intelligibly</li> </ul> |
| <b>4</b> | <ul style="list-style-type: none"> <li>• Life-threatening Consequences of Grade 3 Toxicities</li> <li>• Urgent Intervention Indicated</li> <li>• Mechanical Ventilation</li> </ul>                                                                                                                                                             |
| <b>5</b> | <ul style="list-style-type: none"> <li>• Death</li> </ul>                                                                                                                                                                                                                                                                                      |

\*ADL: activities of daily living

## 19. REFERENCES

1. D'Angio GJ, Evans AE, Koop CE. Special pattern of widespread neuroblastoma with a favourable prognosis. *Lancet*. 1971;1(7708):1046-9. PubMed PMID: 4102970.
2. Evans AE, Baum E, Chard R. Do infants with stage IV-S neuroblastoma need treatment? *ArchDisChild*. 1981;56(4):271-4.
3. Look AT, Hayes FA, Shuster JJ, Douglass EC, Castleberry RP, Bowman LC, Smith EI, Brodeur GM. Clinical relevance of tumor cell ploidy and N-myc gene amplification in childhood neuroblastoma: a Pediatric Oncology Group study. *J Clin Oncol*. 1991;9(4):581-91. doi: 10.1200/jco.1991.9.4.581. PubMed PMID: 2066755.
4. Pole JG, Casper J, Elfenbein G, Gee A, Gross S, Janssen W, Koch P, Marcus R, Pick T, Shuster J, et al. High-dose chemoradiotherapy supported by marrow infusions for advanced neuroblastoma: a Pediatric Oncology Group study. *J Clin Oncol*. 1991;9(1):152-8. doi: 10.1200/jco.1991.9.1.152. PubMed PMID: 1985165.
5. Anderson JR, Coccia PF. Is more better? Dose intensity in neuroblastoma. *J Clin Oncol*. 1991;9(6):902-4.
6. Shuster JJ, Cantor AB, McWilliams N, Pole JG, Castleberry RP, Marcus R, Pick T, Smith EI, Hayes FA. The prognostic significance of autologous bone marrow transplant in advanced neuroblastoma. *J Clin Oncol*. 1991;9(6):1045-9. doi: 10.1200/jco.1991.9.6.1045. PubMed PMID: 2033418.
7. Pinkerton CR. Where next with therapy in advanced neuroblastoma? *British journal of cancer*. 1990;61(3):351-3. PubMed PMID: 2183870; PMCID: PMC1971292.
8. Moreno L, Vaidya SJ, Pinkerton CR, Lewis IJ, Imeson J, Machin D, Pearson AD, European Neuroblastoma Study G, Children's C, Leukaemia G. Long-term follow-up of children with high-risk neuroblastoma: the ENSG5 trial experience. *Pediatr Blood Cancer*. 2013;60(7):1135-40. doi: 10.1002/pbc.24452. PubMed PMID: 23281263.
9. Nasr S, McKolanis J, Pais R, Findley H, Hnath R, Waldrep K, Ragab AH. A phase I study of interleukin-2 in children with cancer and evaluation of clinical and immunologic status during therapy. A Pediatric Oncology Group Study. *Cancer*. 1989;64(4):783-8. PubMed PMID: 2787201.
10. Negrier S, Michon J, Floret D, Bouffet E, Gentet JC, Philip I, Cochat P, Stamm D, Costil J, Gaspard M, et al. Interleukin-2 and lymphokine-activated killer cells in 15 children with advanced metastatic neuroblastoma. *J Clin Oncol*. 1991;9(8):1363-70. doi: 10.1200/jco.1991.9.8.1363. PubMed PMID: 1830096.
11. Cheung NK. Immunotherapy. Neuroblastoma as a model. *Pediatr Clin North Am*. 1991;38(2):425-41.
12. Zhao XJ, Cheung NK. GD2 oligosaccharide: target for cytotoxic T lymphocytes. *The Journal of experimental medicine*. 1995;182(1):67-74. PubMed PMID: 7540657; PMCID: PMC2192096.
13. Foreman NK, Rill DR, Coustan-Smith E, Douglass EC, Brenner MK. Mechanisms of selective killing of neuroblastoma cells by natural killer cells and lymphokine activated killer cells. Potential for residual disease eradication. *British journal of cancer*. 1993;67(5):933-8. PubMed PMID: 8494726; PMCID: PMC1968443.
14. Ucar K, Seeger RC, Challita PM, Watanabe CT, Yen TL, Morgan JP, Amado R, Chou E, McCallister T, Barber JR, et al. Sustained cytokine production and immunophenotypic changes in human neuroblastoma cell lines transduced with a human gamma interferon vector. *Cancer Gene Ther*. 1995;2(3):171-81. PubMed PMID: 8528960.

15. Hank JA, Robinson RR, Surfus J, Mueller BM, Reisfeld RA, Cheung NK, Sondel PM. Augmentation of antibody dependent cell mediated cytotoxicity following in vivo therapy with recombinant interleukin 2. *Cancer Res.* 1990;50(17):5234-9. PubMed PMID: 2386933.
16. Handgretinger R, Bruchelt G, Daurer B, Lang P, Dopfer R, Reisfeld RA, Treuner J, Niethammer D. [The role of interferons in neuroblastoma. 2: Immunomodulatory effects]. *Klin Padiatr.* 1990;202(4):206-11. doi: 10.1055/s-2007-1025521. PubMed PMID: 2118581.
17. Watanabe Y, Kuribayashi K, Miyatake S, Nishihara K, Nakayama E, Taniyama T, Sakata T. Exogenous expression of mouse interferon gamma cDNA in mouse neuroblastoma C1300 cells results in reduced tumorigenicity by augmented anti-tumor immunity. *Proc Natl Acad Sci U S A.* 1989;86(23):9456-60. PubMed PMID: 2512580; PMCID: PMC298515.
18. Bowman L, Grossmann M, Rill D, Brown M, Zhong WY, Alexander B, Leimig T, Coustan-Smith E, Campana D, Jenkins J, Woods D, Kitchingman G, Vanin E, Brenner M. IL-2 adenovector-transduced autologous tumor cells induce antitumor immune responses in patients with neuroblastoma. *Blood.* 1998;92(6):1941-9. PubMed PMID: 9731051.
19. Bowman LC, Grossmann M, Rill D, Brown M, Zhong WY, Alexander B, Leimig T, Coustan-Smith E, Campana D, Jenkins J, Woods D, Brenner M. Interleukin-2 gene-modified allogeneic tumor cells for treatment of relapsed neuroblastoma. *Hum Gene Ther.* 1998;9(9):1303-11. doi: 10.1089/hum.1998.9.9-1303. PubMed PMID: 9650615.
20. Brenner MK, Heslop H, Krance R, Horowitz M, Strother D, Nuchtern J, Grilley B, Martingano E, Cooper K. Phase I study of chemokine and cytokine gene-modified autologous neuroblastoma cells for treatment of relapsed/refractory neuroblastoma using an adenoviral vector. *HumGene Ther.* 2000;11(10):1477-88.
21. Yu AL, Gilman AL, Ozkaynak MF, London WB, Kreissman SG, Chen HX, Smith M, Anderson B, Villablanca JG, Matthay KK, Shimada H, Grupp SA, Seeger R, Reynolds CP, Buxton A, Reisfeld RA, Gillies SD, Cohn SL, Maris JM, Sondel PM, Children's Oncology G. Anti-GD2 antibody with GM-CSF, interleukin-2, and isotretinoin for neuroblastoma. *N Engl J Med.* 2010;363(14):1324-34. doi: 10.1056/NEJMoa0911123. PubMed PMID: 20879881; PMCID: PMC3086629.
22. Yu AL, Uttenreuther-Fischer MM, Huang CS, Tsui CC, Gillies SD, Reisfeld RA, Kung FH. Phase I trial of a human-mouse chimeric anti-disialoganglioside monoclonal antibody ch14.18 in patients with refractory neuroblastoma and osteosarcoma. *J Clin Oncol.* 1998;16(6):2169-80. doi: 10.1200/jco.1998.16.6.2169. PubMed PMID: 9626218.
23. Cheung NK, Kushner BH, Cheung IY, Kramer K, Canete A, Gerald W, Bonilla MA, Finn R, Yeh SJ, Larson SM. Anti-G(D2) antibody treatment of minimal residual stage 4 neuroblastoma diagnosed at more than 1 year of age. *JClinOncol.* 1998;16(9):3053-60.
24. Handgretinger R, Anderson K, Lang P, Dopfer R, Klingebiel T, Schrappe M, Reuland P, Gillies SD, Reisfeld RA, Neithammer D. A phase I study of human/mouse chimeric antiganglioside GD2 antibody ch14.18 in patients with neuroblastoma. *Eur J Cancer.* 1995;31A(2):261-7. PubMed PMID: 7718335.
25. Frost JD, Hank JA, Reaman GH, Friedrich S, Seeger RC, Gan J, Anderson PM, Ettinger LJ, Cairo MS, Blazar BR, Krailo MD, Matthay KK, Reisfeld RA, Sondel PM. A phase I/IB trial of murine monoclonal anti-GD2 antibody 14.G2a plus interleukin-2 in children with refractory neuroblastoma: a report of the Children's Cancer Group. *Cancer.* 1997;80(2):317-33. PubMed PMID: 9217046.
26. Mody R, Naranjo A, Van Ryn C, Yu AL, London WB, Shulkin BL, Parisi MT, Servaes SE, Diccianni MB, Sondel PM, Bender JG, Maris JM, Park JR, Bagatell R. Irinotecan-temozolomide with temsirolimus or dinutuximab in children with refractory or relapsed neuroblastoma (COG ANBL1221): an open-label, randomised, phase 2 trial. *The Lancet Oncology.* 2017;18(7):946-

57. Epub 2017/05/28. doi: 10.1016/s1470-2045(17)30355-8. PubMed PMID: 28549783; PMCID: PMC5527694.
27. Hwu P, Shafer GE, Treisman J, Schindler DG, Gross G, Cowherd R, Rosenberg SA, Eshhar Z. Lysis of ovarian cancer cells by human lymphocytes redirected with a chimeric gene composed of an antibody variable region and the Fc receptor gamma chain. *The Journal of experimental medicine*. 1993;178(1):361-6. PubMed PMID: 8315392; PMCID: PMC2191075.
28. Romeo C, Seed B. Cellular immunity to HIV activated by CD4 fused to T cell or Fc receptor polypeptides. *Cell*. 1991;64(5):1037-46. PubMed PMID: 1900456.
29. Irving BA, Weiss A. The cytoplasmic domain of the T cell receptor zeta chain is sufficient to couple to receptor-associated signal transduction pathways. *Cell*. 1991;64(5):891-901.
30. Letourneur F, Klausner RD. T-cell and basophil activation through the cytoplasmic tail of T-cell-receptor zeta family proteins. *ProcNatlAcadSciUSA*. 1991;88(20):8905-9.
31. Weijtens ME, Willemsen RA, Valerio D, Stam K, Bolhuis RL. Single chain Ig/gamma gene-redredirected human T lymphocytes produce cytokines, specifically lyse tumor cells, and recycle lytic capacity. *Journal of immunology (Baltimore, Md : 1950)*. 1996;157(2):836-43. PubMed PMID: 8752936.
32. Dotti G, Gottschalk S, Savoldo B, Brenner MK. Design and development of therapies using chimeric antigen receptor-expressing T cells. *ImmunolRev*. 2014;257(1):107-26. doi: 10.1111/imr.12131 [doi].
33. Pule MA, Savoldo B, Myers GD, Rossig C, Russell HV, Dotti G, Huls MH, Liu E, Gee AP, Mei Z, Yvon E, Weiss HL, Liu H, Rooney CM, Heslop HE, Brenner MK. Virus-specific T cells engineered to coexpress tumor-specific receptors: persistence and antitumor activity in individuals with neuroblastoma. *Nature medicine*. 2008;14(11):1264-70. doi: 10.1038/nm.1882. PubMed PMID: 18978797; PMCID: PMC2749734.
34. Louis CU, Savoldo B, Dotti G, Pule M, Yvon E, Myers GD, Rossig C, Russell HV, Diouf O, Liu E, Liu H, Wu MF, Gee AP, Mei Z, Rooney CM, Heslop HE, Brenner MK. Antitumor activity and long-term fate of chimeric antigen receptor-positive T cells in patients with neuroblastoma. *Blood*. 2011;118(23):6050-6. doi: 10.1182/blood-2011-05-354449. PubMed PMID: 21984804; PMCID: PMC3234664.
35. Swann JB, Coquet JM, Smyth MJ, Godfrey DI. CD1-restricted T cells and tumor immunity. *Curr Top Microbiol Immunol*. 2007;314:293-323. PubMed PMID: 17593666.
36. Metelitsa LS, Wu HW, Wang H, Yang Y, Warsi Z, Asgharzadeh S, Groshen S, Wilson SB, Seeger RC. Natural killer T cells infiltrate neuroblastomas expressing the chemokine CCL2. *The Journal of experimental medicine*. 2004;199(9):1213-21. doi: 10.1084/jem.20031462. PubMed PMID: 15123743; PMCID: PMC2211904.
37. Tachibana T, Onodera H, Tsuruyama T, Mori A, Nagayama S, Hiai H, Imamura M. Increased intratumor Valpha24-positive natural killer T cells: a prognostic factor for primary colorectal carcinomas. *Clinical cancer research : an official journal of the American Association for Cancer Research*. 2005;11(20):7322-7. doi: 10.1158/1078-0432.CCR-05-0877. PubMed PMID: 16243803.
38. Molling JW, Langius JA, Langendijk JA, Leemans CR, Bontkes HJ, van der Vliet HJ, von Blomberg BM, Scheper RJ, van den Eertwegh AJ. Low levels of circulating invariant natural killer T cells predict poor clinical outcome in patients with head and neck squamous cell carcinoma. *J Clin Oncol*. 2007;25(7):862-8. doi: 10.1200/JCO.2006.08.5787. PubMed PMID: 17327607.
39. Dhodapkar MV. Harnessing human CD1d restricted T cells for tumor immunity: progress and challenges. *Front Biosci (Landmark Ed)*. 2009;14:796-807. PubMed PMID: 19273100; PMCID: PMC2762346.

40. Tahir SM, Cheng O, Shaulov A, Koezuka Y, Bubley GJ, Wilson SB, Balk SP, Exley MA. Loss of IFN-gamma production by invariant NK T cells in advanced cancer. *Journal of immunology* (Baltimore, Md : 1950). 2001;167(7):4046-50. PubMed PMID: 11564825.
41. Fujii S, Shimizu K, Klimek V, Geller MD, Nimer SD, Dhodapkar MV. Severe and selective deficiency of interferon-gamma-producing invariant natural killer T cells in patients with myelodysplastic syndromes. *Br J Haematol*. 2003;122(4):617-22. PubMed PMID: 12899717.
42. Song L, Asgharzadeh S, Salo J, Engell K, Wu HW, Sposto R, Ara T, Silverman AM, DeClerck YA, Seeger RC, Metelitsa LS. Valpha24-invariant NKT cells mediate antitumor activity via killing of tumor-associated macrophages. *J Clin Invest*. 2009;119(6):1524-36. doi: 10.1172/JCI37869. PubMed PMID: 19411762; PMCID: PMC2689106.
43. Liu D, Song L, Wei J, Courtney AN, Gao X, Marinova E, Guo L, Heczey A, Asgharzadeh S, Kim E, Dotti G, Metelitsa LS. IL-15 protects NKT cells from inhibition by tumor-associated macrophages and enhances antimetastatic activity. *J Clin Invest*. 2012;122(6):2221-33. doi: 10.1172/JCI59535. PubMed PMID: 22565311; PMCID: PMC3366399.
44. Giaccone G, Punt CJ, Ando Y, Ruijter R, Nishi N, Peters M, von Blomberg BM, Scheper RJ, van der Vliet HJ, van den Eertwegh AJ, Roelvink M, Beijnen J, Zwierzina H, Pinedo HM. A phase I study of the natural killer T-cell ligand alpha-galactosylceramide (KRN7000) in patients with solid tumors. *Clinical cancer research : an official journal of the American Association for Cancer Research*. 2002;8(12):3702-9. PubMed PMID: 12473579.
45. Nieda M, Okai M, Tazbirkova A, Lin H, Yamaura A, Ide K, Abraham R, Juji T, Macfarlane DJ, Nicol AJ. Therapeutic activation of Valpha24+Vbeta11+ NKT cells in human subjects results in highly coordinated secondary activation of acquired and innate immunity. *Blood*. 2004;103(2):383-9. doi: 10.1182/blood-2003-04-1155. PubMed PMID: 14512316.
46. Ishikawa A, Motohashi S, Ishikawa E, Fuchida H, Higashino K, Otsuji M, Iizasa T, Nakayama T, Taniguchi M, Fujisawa T. A phase I study of alpha-galactosylceramide (KRN7000)-pulsed dendritic cells in patients with advanced and recurrent non-small cell lung cancer. *Clinical cancer research : an official journal of the American Association for Cancer Research*. 2005;11(5):1910-7. doi: 10.1158/1078-0432.CCR-04-1453. PubMed PMID: 15756017.
47. Chang DH, Osman K, Connolly J, Kukreja A, Krasovsky J, Pack M, Hutchinson A, Geller M, Liu N, Annable R, Shay J, Kirchhoff K, Nishi N, Ando Y, Hayashi K, Hassoun H, Steinman RM, Dhodapkar MV. Sustained expansion of NKT cells and antigen-specific T cells after injection of alpha-galactosyl-ceramide loaded mature dendritic cells in cancer patients. *The Journal of experimental medicine*. 2005;201(9):1503-17. doi: 10.1084/jem.20042592. PubMed PMID: 15867097; PMCID: PMC1389847.
48. Motohashi S, Nagato K, Kunii N, Yamamoto H, Yamasaki K, Okita K, Hanaoka H, Shimizu N, Suzuki M, Yoshino I, Taniguchi M, Fujisawa T, Nakayama T. A phase I-II study of alpha-galactosylceramide-pulsed IL-2/GM-CSF-cultured peripheral blood mononuclear cells in patients with advanced and recurrent non-small cell lung cancer. *Journal of immunology* (Baltimore, Md : 1950). 2009;182(4):2492-501. doi: 10.4049/jimmunol.0800126. PubMed PMID: 19201905.
49. Kunii N, Horiguchi S, Motohashi S, Yamamoto H, Ueno N, Yamamoto S, Sakurai D, Taniguchi M, Nakayama T, Okamoto Y. Combination therapy of in vitro-expanded natural killer T cells and alpha-galactosylceramide-pulsed antigen-presenting cells in patients with recurrent head and neck carcinoma. *Cancer Sci*. 2009;100(6):1092-8. doi: 10.1111/j.1349-7006.2009.01135.x. PubMed PMID: 19302288.
50. Yamasaki K, Horiguchi S, Kurosaki M, Kunii N, Nagato K, Hanaoka H, Shimizu N, Ueno N, Yamamoto S, Taniguchi M, Motohashi S, Nakayama T, Okamoto Y. Induction of NKT cell-specific immune responses in cancer tissues after NKT cell-targeted adoptive immunotherapy. *Clin Immunol*. 2011;138(3):255-65. doi: 10.1016/j.clim.2010.11.014. PubMed PMID: 21185787.

51. Exley MA, Friedlander P, Alatrakchi N, Vriend L, Yue SC, Sasada T, Zang W, Mizukami Y, Clark J, Nemer D, LeClair K, Canning C, Daley H, Dranoff G, Giobbie-Hurder A, Hodi FS, Ritz J, Balk SP. Adoptive Transfer of Invariant NKT Cells as Immunotherapy for Advanced Melanoma: a Phase 1 Clinical Trial. *Clinical cancer research : an official journal of the American Association for Cancer Research*. 2017. doi: 10.1158/1078-0432.CCR-16-0600. PubMed PMID: 28193627.
52. Kronenberg M, Gapin L. The unconventional lifestyle of NKT cells. *Nat Rev Immunol*. 2002;2(8):557-68. doi: 10.1038/nri854. PubMed PMID: 12154375.
53. van den Heuvel MJ, Garg N, Van Kaer L, Haeryfar SM. NKT cell costimulation: experimental progress and therapeutic promise. *Trends Mol Med*. 2011;17(2):65-77. doi: 10.1016/j.molmed.2010.10.007. PubMed PMID: 21087900; PMCID: PMC3616392.
54. Kawano T, Cui J, Koezuka Y, Toura I, Kaneko Y, Motoki K, Ueno H, Nakagawa R, Sato H, Kondo E, Koseki H, Taniguchi M. CD1d-restricted and TCR-mediated activation of valpha14 NKT cells by glycosylceramides. *Science*. 1997;278(5343):1626-9. PubMed PMID: 9374463.
55. Hayakawa Y, Takeda K, Yagita H, Van Kaer L, Saiki I, Okumura K. Differential regulation of Th1 and Th2 functions of NKT cells by CD28 and CD40 costimulatory pathways. *Journal of immunology (Baltimore, Md : 1950)*. 2001;166(10):6012-8. PubMed PMID: 11342617.
56. DelaRosa O, Tarazona R, Casado JG, Alonso C, Ostos B, Pena J, Solana R. Valpha24+ NKT cells are decreased in elderly humans. *Exp Gerontol*. 2002;37(2-3):213-7. PubMed PMID: 11772506.
57. Cole SL, Benam KH, McMichael AJ, Ho LP. Involvement of the 4-1BB/4-1BBL pathway in control of monocyte numbers by invariant NKT cells. *Journal of immunology (Baltimore, Md : 1950)*. 2014;192(8):3898-907. Epub 2014/03/19. doi: 10.4049/jimmunol.1302385. PubMed PMID: 24639347.
58. Kobayashi T, Doff BL, Rearden RC, Leggatt GR, Mattarollo SR. NKT cell-targeted vaccination plus anti-4-1BB antibody generates persistent CD8 T cell immunity against B cell lymphoma. *Oncoimmunology*. 2015;4(3):e990793. Epub 2015/05/08. doi: 10.4161/2162402x.2014.990793. PubMed PMID: 25949907; PMCID: PMC4404843.
59. Boesteanu AC, Katsikis PD. Memory T cells need CD28 costimulation to remember. *Seminars in immunology*. 2009;21(2):69-77. Epub 2009/03/10. doi: 10.1016/j.smim.2009.02.005. PubMed PMID: 19268606; PMCID: PMC2923542.
60. Sharpe AH, Freeman GJ. The B7-CD28 superfamily. *Nat Rev Immunol*. 2002;2(2):116-26. doi: 10.1038/nri727. PubMed PMID: 11910893.
61. Gimmi CD, Freeman GJ, Gribben JG, Gray G, Nadler LM. Human T-cell clonal anergy is induced by antigen presentation in the absence of B7 costimulation. *Proc Natl Acad Sci USA*. 1993;90(14):6586-90.
62. Ramos CA, Dotti G. Chimeric antigen receptor (CAR)-engineered lymphocytes for cancer therapy. *Expert Opin Biol Ther*. 2011;11(7):855-73. doi: 10.1517/14712598.2011.573476. PubMed PMID: 21463133; PMCID: PMC3107373.
63. Zhang L, Morgan RA, Beane JD, Zheng Z, Dudley ME, Kassim SH, Nahvi AV, Ngo LT, Sherry RM, Phan GQ, Hughes MS, Kammula US, Feldman SA, Toomey MA, Kerkar SP, Restifo NP, Yang JC, Rosenberg SA. Tumor-infiltrating lymphocytes genetically engineered with an inducible gene encoding interleukin-12 for the immunotherapy of metastatic melanoma. *Clinical cancer research : an official journal of the American Association for Cancer Research*. 2015;21(10):2278-88. doi: 10.1158/1078-0432.CCR-14-2085. PubMed PMID: 25695689; PMCID: 4433819.
64. Sim GC, Radvanyi L. The IL-2 cytokine family in cancer immunotherapy. *Cytokine & growth factor reviews*. 2014;25(4):377-90. Epub 2014/09/10. doi: 10.1016/j.cytogfr.2014.07.018. PubMed PMID: 25200249.

65. Hoyos V, Savoldo B, Quintarelli C, Mahendravada A, Zhang M, Vera J, Heslop HE, Rooney CM, Brenner MK, Dotti G. Engineering CD19-specific T lymphocytes with interleukin-15 and a suicide gene to enhance their anti-lymphoma/leukemia effects and safety. *Leukemia*. 2010;24(6):1160-70. doi: leu201075 [pii];10.1038/leu.2010.75 [doi].
66. Markley JC, Sadelain M. IL-7 and IL-21 are superior to IL-2 and IL-15 in promoting human T cell-mediated rejection of systemic lymphoma in immunodeficient mice. *Blood*. 2010;115(17):3508-19. doi: 10.1182/blood-2009-09-241398. PubMed PMID: 20190192; PMCID: PMC2867264.
67. Long AH, Haso WM, Shern JF, Wanhainen KM, Murgai M, Ingaramo M, Smith JP, Walker AJ, Kohler ME, Venkateshwara VR, Kaplan RN, Patterson GH, Fry TJ, Orentas RJ, Mackall CL. 4-1BB costimulation ameliorates T cell exhaustion induced by tonic signaling of chimeric antigen receptors. *Nature medicine*. 2015;21(6):581-90. doi: 10.1038/nm.3838. PubMed PMID: 25939063; PMCID: PMC4458184.
68. Heczey A, Liu D, Tian G, Courtney AN, Wei J, Marinova E, Gao X, Guo L, Yvon E, Hicks J, Liu H, Dotti G, Metelitsa LS. Invariant NKT cells with chimeric antigen receptor provide a novel platform for safe and effective cancer immunotherapy. *Blood*. 2014;124(18):2824-33. doi: 10.1182/blood-2013-11-541235. PubMed PMID: 25049283; PMCID: PMC4215313.
69. Maude SL, Barrett D, Teachey DT, Grupp SA. Managing cytokine release syndrome associated with novel T cell-engaging therapies. *Cancer journal (Sudbury, Mass)*. 2014;20(2):119-22. doi: 10.1097/PPO.000000000000035. PubMed PMID: 24667956; PMCID: PMC4119809.
70. Maude SL, Frey N, Shaw PA, Aplenc R, Barrett DM, Bunin NJ, Chew A, Gonzalez VE, Zheng Z, Lacey SF, Mahnke YD, Melenhorst JJ, Rheingold SR, Shen A, Teachey DT, Levine BL, June CH, Porter DL, Grupp SA. Chimeric antigen receptor T cells for sustained remissions in leukemia. *The New England journal of medicine*. 2014;371(16):1507-17. doi: 10.1056/NEJMoa1407222 [doi].
71. Turtle CJ, Hanafi LA, Berger C, Gooley TA, Cherian S, Hudecek M, Sommermeyer D, Melville K, Pender B, Budiarto TM, Robinson E, Steevens NN, Chaney C, Soma L, Chen X, Yeung C, Wood B, Li D, Cao J, Heimfeld S, Jensen MC, Riddell SR, Maloney DG. CD19 CAR-T cells of defined CD4+:CD8+ composition in adult B cell ALL patients. *J Clin Invest*. 2016;126(6):2123-38. doi: 10.1172/JCI85309. PubMed PMID: 27111235; PMCID: PMC4887159.
72. Kochenderfer JN, Dudley ME, Feldman SA, Wilson WH, Spaner DE, Maric I, Stetler-Stevenson M, Phan GQ, Hughes MS, Sherry RM, Yang JC, Kammula US, Devillier L, Carpenter R, Nathan DA, Morgan RA, Laurencot C, Rosenberg SA. B-cell depletion and remissions of malignancy along with cytokine-associated toxicity in a clinical trial of anti-CD19 chimeric-antigen-receptor-transduced T cells. *Blood*. 2012;119(12):2709-20. doi: 10.1182/blood-2011-10-384388. PubMed PMID: 22160384; PMCID: 3327450.
73. Lee DW, Kochenderfer JN, Stetler-Stevenson M, Cui YK, Delbrook C, Feldman SA, Fry TJ, Orentas R, Sabatino M, Shah NN, Steinberg SM, Stroncek D, Tschernia N, Yuan C, Zhang H, Zhang L, Rosenberg SA, Wayne AS, Mackall CL. T cells expressing CD19 chimeric antigen receptors for acute lymphoblastic leukaemia in children and young adults: a phase 1 dose-escalation trial. *The Lancet*. 385(9967):517-28. doi: 10.1016/S0140-6736(14)61403-3.
74. Fehniger TA, Suzuki K, Ponnappan A, VanDeusen JB, Cooper MA, Florea SM, Freud AG, Robinson ML, Durbin J, Caligiuri MA. Fatal leukemia in interleukin 15 transgenic mice follows early expansions in natural killer and memory phenotype CD8+ T cells. *The Journal of experimental medicine*. 2001;193(2):219-31. Epub 2001/02/24. PubMed PMID: 11208862; PMCID: PMC2193336.

75. Fehniger TA, Suzuki K, VanDeusen JB, Cooper MA, Freud AG, Caligiuri MA. Fatal leukemia in interleukin-15 transgenic mice. *Blood cells, molecules & diseases*. 2001;27(1):223-30. Epub 2001/05/19. doi: 10.1006/bcmd.2001.0379. PubMed PMID: 11358383.
76. Hsu C, Jones SA, Cohen CJ, Zheng Z, Kerstann K, Zhou J, Robbins PF, Peng PD, Shen X, Gomes TJ, Dunbar CE, Munroe DJ, Stewart C, Cornetta K, Wangsa D, Ried T, Rosenberg SA, Morgan RA. Cytokine-independent growth and clonal expansion of a primary human CD8+ T-cell clone following retroviral transduction with the IL-15 gene. *Blood*. 2007;109(12):5168-77. Epub 2007/03/14. doi: 10.1182/blood-2006-06-029173. PubMed PMID: 17353346; PMCID: PMC1890824.
77. Conlon KC, Lugli E, Welles HC, Rosenberg SA, Fojo AT, Morris JC, Fleisher TA, Dubois SP, Perera LP, Stewart DM, Goldman CK, Bryant BR, Decker JM, Chen J, Worthy TA, Figg WD, Sr., Peer CJ, Sneller MC, Lane HC, Yovandich JL, Creekmore SP, Roederer M, Waldmann TA. Redistribution, hyperproliferation, activation of natural killer cells and CD8 T cells, and cytokine production during first-in-human clinical trial of recombinant human interleukin-15 in patients with cancer. *J Clin Oncol*. 2015;33(1):74-82. Epub 2014/11/19. doi: 10.1200/jco.2014.57.3329. PubMed PMID: 25403209; PMCID: PMC4268254.
78. Song L, Ara T, Wu HW, Woo CW, Reynolds CP, Seeger RC, DeClerck YA, Thiele CJ, Sposto R, Metelitsa LS. Oncogene MYCN regulates localization of NKT cells to the site of disease in neuroblastoma. *J Clin Invest*. 2007;117(9):2702-12. doi: 10.1172/JCI30751. PubMed PMID: 17710228; PMCID: PMC1940236.
79. Asgharzadeh S, Salo JA, Ji L, Oberthuer A, Fischer M, Berthold F, Hadjidaniel M, Liu CW, Metelitsa LS, Pique-Regi R, Wakamatsu P, Villablanca JG, Kreissman SG, Matthay KK, Shimada H, London WB, Sposto R, Seeger RC. Clinical significance of tumor-associated inflammatory cells in metastatic neuroblastoma. *J Clin Oncol*. 2012;30(28):3525-32. Epub 2012/08/29. doi: 10.1200/JCO.2011.40.9169. PubMed PMID: 22927533; PMCID: PMC3675667.
80. Muranski P, Boni A, Wrzesinski C, Citrin DE, Rosenberg SA, Childs R, Restifo NP. Increased intensity lymphodepletion and adoptive immunotherapy--how far can we go? *NatClinPractOncol*. 2006;3(12):668-81. doi: ncponc0666 [pii];10.1038/ncponc0666 [doi].
81. Muranski P, Chmielowski B, Ignatowicz L. Mature CD4+ T cells perceive a positively selecting class II MHC/peptide complex in the periphery. *Journal of immunology (Baltimore, Md : 1950)*. 2000;164(6):3087-94. doi: ji\_v164n6p3087 [pii]. PubMed PMID: 10706698.
82. Kirberg J, Berns A, von Boehmer H. Peripheral T cell survival requires continual ligation of the T cell receptor to major histocompatibility complex-encoded molecules. *The Journal of experimental medicine*. 1997;186(8):1269-75. PubMed PMID: 9334366; PMCID: PMC2199081.
83. Kondrack RM, Harbertson J, Tan JT, McBreen ME, Surh CD, Bradley LM. Interleukin 7 regulates the survival and generation of memory CD4 cells. *The Journal of experimental medicine*. 2003;198(12):1797-806. doi: 10.1084/jem.20030735. PubMed PMID: 14662907; PMCID: PMC2194153.
84. Ku CC, Murakami M, Sakamoto A, Kappler J, Marrack P. Control of homeostasis of CD8+ memory T cells by opposing cytokines. *Science*. 2000;288(5466):675-8. doi: 8464 [pii]. PubMed PMID: 10784451.
85. Schluns KS, Kieper WC, Jameson SC, Lefrancois L. Interleukin-7 mediates the homeostasis of naive and memory CD8 T cells in vivo. *Nature immunology*. 2000;1(5):426-32. doi: 10.1038/80868. PubMed PMID: 11062503.
86. Goldrath AW, Sivakumar PV, Glaccum M, Kennedy MK, Bevan MJ, Benoist C, Mathis D, Butz EA. Cytokine requirements for acute and Basal homeostatic proliferation of naive and memory CD8+ T cells. *The Journal of experimental medicine*. 2002;195(12):1515-22. PubMed PMID: 12070279; PMCID: PMC2193554.

87. Zeng R, Spolski R, Finkelstein SE, Oh S, Kovanen PE, Hinrichs CS, Pise-Masison CA, Radonovich MF, Brady JN, Restifo NP, Berzofsky JA, Leonard WJ. Synergy of IL-21 and IL-15 in regulating CD8+ T cell expansion and function. *The Journal of experimental medicine*. 2005;201(1):139-48. doi: 10.1084/jem.20041057. PubMed PMID: 15630141; PMCID: PMC2212766.
88. Sakaguchi S, Fukuma K, Kuribayashi K, Masuda T. Organ-specific autoimmune diseases induced in mice by elimination of T cell subset. I. Evidence for the active participation of T cells in natural self-tolerance; deficit of a T cell subset as a possible cause of autoimmune disease. *The Journal of experimental medicine*. 1985;161(1):72-87. PubMed PMID: 3871469; PMCID: PMC2187548.
89. Gattinoni L, Finkelstein SE, Klebanoff CA, Antony PA, Palmer DC, Spiess PJ, Hwang LN, Yu Z, Wrzesinski C, Heimann DM, Surh CD, Rosenberg SA, Restifo NP. Removal of homeostatic cytokine sinks by lymphodepletion enhances the efficacy of adoptively transferred tumor-specific CD8+ T cells. *The Journal of experimental medicine*. 2005;202(7):907-12. doi: 10.1084/jem.20050732. PubMed PMID: 16203864; PMCID: PMC1397916.
90. Dudley ME, Wunderlich J, Nishimura MI, Yu D, Yang JC, Topalian SL, Schwartzentruber DJ, Hwu P, Marincola FM, Sherry R, Leitman SF, Rosenberg SA. Adoptive transfer of cloned melanoma-reactive T lymphocytes for the treatment of patients with metastatic melanoma. *Journal of immunotherapy*. 2001;24(4):363-73. PubMed PMID: 11565838.
91. Dudley ME, Yang JC, Sherry R, Hughes MS, Royal R, Kammula U, Robbins PF, Huang J, Citrin DE, Leitman SF, Wunderlich J, Restifo NP, Thomasian A, Downey SG, Smith FO, Klapper J, Morton K, Laurencot C, White DE, Rosenberg SA. Adoptive cell therapy for patients with metastatic melanoma: evaluation of intensive myeloablative chemoradiation preparative regimens. *J Clin Oncol*. 2008;26(32):5233-9. doi: 10.1200/JCO.2008.16.5449. PubMed PMID: 18809613; PMCID: PMC2652090.
92. Grupp SA, Kalos M, Barrett D, Aplenc R, Porter DL, Rheingold SR, Teachey DT, Chew A, Hauck B, Wright JF, Milone MC, Levine BL, June CH. Chimeric antigen receptor-modified T cells for acute lymphoid leukemia. *N Engl J Med*. 2013;368(16):1509-18. doi: 10.1056/NEJMoa1215134. PubMed PMID: 23527958; PMCID: PMC4058440.
93. Davila ML, Riviere I, Wang X, Bartido S, Park J, Curran K, Chung SS, Stefanski J, Borquez-Ojeda O, Olszewska M, Qu J, Wasielewska T, He Q, Fink M, Shinglot H, Youssif M, Satter M, Wang Y, Hosey J, Quintanilla H, Halton E, Bernal Y, Bouhassira DC, Arcila ME, Gonen M, Roboz GJ, Maslak P, Douer D, Frattini MG, Giralto S, Sadelain M, Brentjens R. Efficacy and toxicity management of 19-28z CAR T cell therapy in B cell acute lymphoblastic leukemia. *Science translational medicine*. 2014;6(224):224ra25. doi: 10.1126/scitranslmed.3008226. PubMed PMID: 24553386; PMCID: PMC4684949.
94. Scholler J, Brady TL, Binder-Scholl G, Hwang WT, Plesa G, Hege KM, Vogel AN, Kalos M, Riley JL, Deeks SG, Mitsuyasu RT, Bernstein WB, Aronson NE, Levine BL, Bushman FD, June CH. Decade-long safety and function of retroviral-modified chimeric antigen receptor T cells. *Science translational medicine*. 2012;4(132):132ra53. doi: 10.1126/scitranslmed.3003761. PubMed PMID: 22553251; PMCID: PMC4368443.
95. Porter DL, Levine BL, Kalos M, Bagg A, June CH. Chimeric antigen receptor-modified T cells in chronic lymphoid leukemia. *N Engl J Med*. 2011;365(8):725-33. doi: 10.1056/NEJMoa1103849. PubMed PMID: 21830940; PMCID: PMC3387277.
96. Pule MA, Savoldo B, Myers GD, Rossig C, Russell hv, Dotti G, Huls MH, Liu E, Gee AP, Mei Z, Yvon E, Weiss HL, Liu H, Rooney CM, Heslop HE, Brenner MK. Virus-specific T cells engineered to coexpress tumor-specific receptors: persistence and antitumor activity in individuals with neuroblastoma. *NatMed*. 2008;14(11):1264-70.

97. Louis CU, Savoldo B, Dotti G, Pule M, Yvon E, Myers GD, Rossig C, Russell hv, Diouf O, Liu E, Liu H, Wu MF, Gee AP, Mei Z, Rooney CM, Heslop HE, Brenner MK. Antitumor activity and long-term fate of chimeric antigen receptor-positive T cells in patients with neuroblastoma. *Blood*. 2011;118(23):6050-6.
98. Lee DW, Gardner R, Porter DL, Louis CU, Ahmed N, Jensen M, Grupp SA, Mackall CL. Current concepts in the diagnosis and management of cytokine release syndrome. *Blood*. 2014;124(2):188-95. doi: 10.1182/blood-2014-05-552729. PubMed PMID: 24876563; PMCID: PMC4093680.
99. Shin DS, Ribas A. The evolution of checkpoint blockade as a cancer therapy: what's here, what's next? *Curr Opin Immunol*. 2015;33:23-35. doi: S0952-7915(15)00007-2 [pii];10.1016/j.coi.2015.01.006 [doi].
100. Teachey DT, Rheingold SR, Maude SL, Zugmaier G, Barrett DM, Seif AE, Nichols KE, Suppa EK, Kalos M, Berg RA, Fitzgerald JC, Aplenc R, Gore L, Grupp SA. Cytokine release syndrome after blinatumomab treatment related to abnormal macrophage activation and ameliorated with cytokine-directed therapy. *Blood*. 2013;121(26):5154-7. doi: 10.1182/blood-2013-02-485623. PubMed PMID: 23678006; PMCID: PMC4123427.
101. Maude SL, Frey N, Shaw PA, Aplenc R, Barrett DM, Bunin NJ, Chew A, Gonzalez VE, Zheng Z, Lacey SF, Mahnke YD, Melenhorst JJ, Rheingold SR, Shen A, Teachey DT, Levine BL, June CH, Porter DL, Grupp SA. Chimeric antigen receptor T cells for sustained remissions in leukemia. *N Engl J Med*. 2014;371(16):1507-17. doi: 10.1056/NEJMoa1407222. PubMed PMID: 25317870; PMCID: PMC4267531.
102. Papadopoulou A, Krance RA, Allen CE, Lee D, Rooney CM, Brenner MK, Leen AM, Heslop HE. Systemic inflammatory response syndrome after administration of unmodified T lymphocytes. *Mol Ther*. 2014;22(6):1134-8. doi: 10.1038/mt.2014.48. PubMed PMID: 24651135; PMCID: PMC4048900.
103. Stevanovic S, Draper LM, Langan MM, Campbell TE, Kwong ML, Wunderlich JR, Dudley ME, Yang JC, Sherry RM, Kammula US, Restifo NP, Rosenberg SA, Hinrichs CS. Complete regression of metastatic cervical cancer after treatment with human papillomavirus-targeted tumor-infiltrating T cells. *J Clin Oncol*. 2015;33(14):1543-50. doi: 10.1200/JCO.2014.58.9093. PubMed PMID: 25823737; PMCID: PMC4417725.
104. Davila ML, Riviere I, Wang X, Bartido S, Park J, Curran K, Chung SS, Stefanski J, Borquez-Ojeda O, Olszewska M, Qu J, Wasielewska T, He Q, Fink M, Shinglot H, Youssif M, Satter M, Wang Y, Hosey J, Quintanilla H, Halton E, Bernal Y, Bouhassira DC, Arcila ME, Gonen M, Roboz GJ, Maslak P, Douer D, Frattini MG, Giralto S, Sadelain M, Brentjens R. Efficacy and Toxicity Management of 19-28z CAR T Cell Therapy in B Cell Acute Lymphoblastic Leukemia. *SciTranslMed*. 2014;6(224):224ra25. doi: 6/224/224ra25 [pii];10.1126/scitranslmed.3008226 [doi].
105. Kochenderfer JN, Dudley ME, Kassim SH, Somerville RP, Carpenter RO, Stetler-Stevenson M, Yang JC, Phan GQ, Hughes MS, Sherry RM, Raffeld M, Feldman S, Lu L, Li YF, Ngo LT, Goy A, Feldman T, Spaner DE, Wang ML, Chen CC, Kranick SM, Nath A, Nathan DA, Morton KE, Toomey MA, Rosenberg SA. Chemotherapy-refractory diffuse large B-cell lymphoma and indolent B-cell malignancies can be effectively treated with autologous T cells expressing an anti-CD19 chimeric antigen receptor. *J Clin Oncol*. 2015;33(6):540-9. doi: JCO.2014.56.2025 [pii];10.1200/JCO.2014.56.2025 [doi].
106. Lee DW, Kochenderfer JN, Stetler-Stevenson M, Cui YK, Delbrook C, Feldman SA, Fry TJ, Orentas R, Sabatino M, Shah NN, Steinberg SM, Stroncek D, Tschernia N, Yuan C, Zhang H, Zhang L, Rosenberg SA, Wayne AS, Mackall CL. T cells expressing CD19 chimeric antigen receptors for acute lymphoblastic leukaemia in children and young adults: a phase 1 dose-

escalation trial. *Lancet*. 2015;385(9967):517-28. doi: 10.1016/S0140-6736(14)61403-3. PubMed PMID: 25319501.

107. Turtle CJ, Hanafi LA, Berger C, Hudecek M, Pender B, Robinson E, Hawkins R, Chaney C, Cherian S, Chen X, Soma L, Wood B, Li D, Heimfeld S, Riddell SR, Maloney DG. Immunotherapy of non-Hodgkin's lymphoma with a defined ratio of CD8+ and CD4+ CD19-specific chimeric antigen receptor-modified T cells. *Science translational medicine*. 2016;8(355):355ra116. doi: 10.1126/scitranslmed.aaf8621. PubMed PMID: 27605551; PMCID: PMC5045301.

108. Metelitsa LS, Naidenko OV, Kant A, Wu HW, Loza MJ, Perussia B, Kronenberg M, Seeger RC. Human NKT cells mediate antitumor cytotoxicity directly by recognizing target cell CD1d with bound ligand or indirectly by producing IL-2 to activate NK cells. *Journal of immunology* (Baltimore, Md : 1950). 2001;167(6):3114-22. PubMed PMID: 11544296.

109. Brodeur GM, Pritchard J, Berthold F, Carlsen NL, Castel V, Castelberry RP, De Bernardi B, Evans AE, Favrot M, Hedborg F, et al. Revisions of the international criteria for neuroblastoma diagnosis, staging, and response to treatment. *J Clin Oncol*. 1993;11(8):1466-77. doi: 10.1200/jco.1993.11.8.1466. PubMed PMID: 8336186.

110. Gray RA. A class of K-sample tests for comparing the cumulative incidence of a competing risk. *Ann Stat*. 1988;16:1141-54.